

Exact Certificate-Fibre Exhaustion and Unique Role-Occupancy Closure under AASC

A Target-Fibred Eight-Sorted Completion Theorem for the Constraint-Native
Gravity–Quantum Capstone

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**Cycle-4 correction: source-locked primary interfaces, exact cosieve completion, and
dependent closure**

Abstract

This manuscript supplies the final certificate-completeness and role-occupancy module required by the AASC gravity–quantum capstone. The dependency order is fixed and irreversible: target adequacy forces the AASC kernel; the kernel governs the realized physical interior; gravity and quantum standing are downstream roles; Einstein and Schrödinger dynamics are later bookkeeping projections; accepted Modules A, B, and C are still later theorem objects. No residue, tensor colimit, skin congruence, assignment, automorphism, or realization conclusion authorizes an upstream structure.

The construction distinguishes generic interface moduli from the exact target-relative fibres of one accepted certificate. Different source systems, potentials, initial data, initial rays, RPI seeds, tensor generator diagrams, or matching packages remain legitimate points of their generic moduli. The frozen antecedent package contains the six non-tensor/non-skin physical target vertices, the exact accepted RPI/AER and Modules A–C tensor generators, their declared common subinterfaces, and their complete probe maps. It does not contain an already completed tensor ledger or an already maximal skin congruence.

The central Cycle-4 repair begins with fifteen source-locked primary generator diagrams and twenty-four source-locked primary overlap relations, none of which is falsely extended by empty target components. Each primary interface is covariantly completed by the explicit left Kan path-term construction along its primary incidence inclusion. The completed supports are the exact forward cosieve closures of the primary supports; every supported source has a supported target and an explicitly registered same-direction incidence map. Each overlap relation is completed by left Kan extension and then replaced by the joint image of its two maps into the adjacent completed generators, so no dummy nonempty component or unsupported target map is introduced. Only after this typed completion is the source-locked diagram assembled and its tensor colimit constructed.

The accepted primary generator probes are extended by the same Kan adjunction, and primary overlap compatibility extends uniquely to each completed joint-image relation. The completed probes form a compatible cocone in the same covariant slice and therefore induce a unique slice-typed tensor probe on the constructed colimit. Combined with the accepted anchor, GR/source, quantum, boundary, transport, matching, failure, and report probes, this yields the frozen joint target-evaluation functor. Its lawful kernel pair is then constructed as the skin target and proved to be the unique maximal standing-neutral congruence. The complete eight-vertex target diagram is therefore formed before candidate presentations, residues, assignments, or orbit calculations.

Candidate responses are generated from the fixed certificate grammar. Generic interface eligibility remains separate from target fidelity. Presentation equivalence is lawful interface isomorphism, not equality under a selected profile. The eight target-relative fibre theorems use, in their exact scopes, the constructed tensor colimit, the dependent skin kernel, fixed-data Cauchy uniqueness, Wigner–Stone classification of one fixed projective process, final record and compatibility quotients, Module C flow uniqueness for one fixed upstream seed, and uniqueness of presentations of the fixed accepted matching cone.

A twenty-five-arrow incidence diagram records all generated cross-coordinate conditions. The accepted Modules A–C presentations give one compatible injection. Exact fibre finality makes every compatible complete injection identical on residue classes; hence the independently formed live residue is exactly the eight accepted classes. The complete assignment space has one lawful orbit. After candidate, quotient, role-set, eligibility, compatibility, live-residue, and failure completeness are separately verified, the imported Role-Occupancy Closure theorem licenses one complete constraint-native realization class for the fixed target. Generic GR, quantum, transport, tensor-diagram, and matching moduli are not claimed singleton.

Dependency lock.

$$\text{TargetAdeq}_D \implies K_D = (\text{Adm}_D, \text{Stand}_D, \text{Ref}_D, \text{Irr}_D) \implies A_D^+ \implies I_D.$$

Only then:

$$\begin{aligned} I_D &\implies \{\text{Gravity}_D, \text{QuantumStand}_D\} \implies \{\text{Einstein}_D, \text{Schrodinger}_D\}, \\ \{\text{Einstein}_D, \text{Schrodinger}_D\} &\implies \text{MA}_{\text{src}} \implies \text{MB}_{\text{mass}} \implies \text{MC}_{\text{cont}} \implies \text{MD}_{\text{occ}}. \end{aligned}$$

Module D is a final target-relative completion theorem for an already constructed certificate. It generates no source, dynamics, state, seed, empirical value, or upstream role.

Reportable standalone endpoint. Relative to the exact accepted Modules A–C antecedent package, the tensor ledger is explicitly constructed as the colimit of the source-locked generator/overlap diagram; the skin object is then constructed as the maximal lawful kernel of the induced complete joint probe. Every certificate-generated target-relative role fibre has one lawful presentation class, the complete incidence diagram is compatible, the live residue formed from all compatible injections is exactly the eight accepted classes, and

$$\left| \Sigma_D^{gg} / \Gamma_D^{\text{lawful}} \right| = 1.$$

Consequently the role-realization-complete certificate satisfies Role-Occupancy Closure and licenses one complete constraint-native realization class for the fixed target. Generic interface moduli are not claimed singleton.

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Part I

Dependency Lock and Frozen Module Target

1 Proof class, exact target instance, and immutable dependency order

The proof class is *target-fibred certificate completion and role-occupancy closure*. Module D does not construct the realized physical interior, a metric source, a common mass charge, an upstream continuation, or a projection cube. Those objects have already been formed and independently reviewed. Module D's new constructional burdens are narrower: assemble the accepted standing-bearing generators into one exact tensor ledger; form the dependent maximal skin congruence; and then decide whether the resulting eight-role certificate is complete and uniquely occupied for one fixed target instance.

Definition 1.1 (Exact frozen target instance). The target T_D^{gq} is determined by:

1. the FEP-1.0 capstone contract;
2. the singleton RPI and its declared regular local comparison envelope;
3. the accepted Module A source-bearing subtype, attachment, source tensor, Ward, and finite composite data;
4. the accepted Module B operator/source attachment, stable scalar charge line, active and inertial readouts, and calibration package;
5. the accepted Module C upstream history, fixed-clock category, lawful metric/source orbit, full-ray quantum process, record and compatibility quotients, directed exchange data, and sixteen-vertex cube;
6. the exact report and failure boundary stated in the frozen endpoint;
7. the source-locked tensor generator objects, common subinterfaces, and complete generator probes derived from the preceding accepted data.

Items 1–7 form the antecedent package $\Theta_D^{ABC,-}$. The tensor target and skin target are not included as imported conclusions; they are constructed from this package in section 5. The complete target diagram Θ_D^{ABC} is fixed only after those constructions and still before candidate inspection.

Remark 1.2 (Exact instance versus generic interface). The broad interface categories remain nontrivial. There are many Einstein–matter systems, self-adjoint Hamiltonians, initial rays, RPI seeds, record profiles, tensor generator diagrams, and matching packages. Module D does not collapse them. It

classifies which presentations realize the exact accepted object constructed from $\Theta_D^{ABC,-}$. The theorem is therefore a completion theorem for one certificate instance, not a uniqueness theorem for an entire interface class.

Definition 1.3 (Correctness-relevant criticism). A criticism is correctness-relevant to Module D only if it identifies one of the following:

1. an upstream datum inserted after candidate inspection;
2. an incomplete or ill-typed source-locked tensor generator/overlap diagram;
3. an invalid tensor-colimit construction or mediator proof;
4. an incomplete joint probe or unsound skin-kernel construction;
5. an incomplete target-relative candidate or comparison category;
6. a presentation quotient that removes target-bearing structure or retains pure presentation multiplicity;
7. a confusion between generic interface eligibility and target equivalence;
8. an incomplete required-incidence registry or false compatibility equation;
9. an invalid fibre theorem, live-residue calculation, automorphism action, or Role-Occupancy invocation;
10. an overreported conclusion beyond the exact target instance.

A demand for different dynamics, a different source, a different initial state, a different clock, or a broader quantum-gravity theory changes the target unless it shows that the allegedly different object is lawfully equivalent to the fixed target data.

Theorem 1.4 (Dependency-inversion exclusion). *No object introduced in Module D may authorize, generate, or repair an upstream object. In particular,*

$$\text{tensor colimit} \not\Rightarrow I_D, \quad \text{one assignment orbit} \not\Rightarrow \text{Gravity}_D,$$

and no certificate comparison map may create source standing, quantum standing, continuation, or record standing.

Proof. The antecedent package is fixed entirely from accepted objects before the tensor and skin constructions. The tensor colimit only assembles those already accepted generator interfaces through declared overlap maps. The skin kernel only quotients lawful presentation differences invisible to already accepted probes. Candidate comparison maps then point from proposed presentations to the fixed target objects. None supplies the physical carriers or theorem objects that the presentations realize. Reversing the order would use certificate bookkeeping as a generator of the structures it audits, contrary to kernel no-generator and no-backflow conditions. $\square$

Remark 1.5 (Why Module D is not redundant). Modules A–C establish that the source, mass-charge, continuation, projection, record, compatibility, and exchange objects exist and satisfy their own endpoints. They do not construct the final lawful tensor colimit, the dependent maximal skin congruence, the complete eight-role incidence certificate, the live residue, or the unique assignment orbit. Those are Module D’s distinct burdens.

2 Accepted antecedent theorem objects and the source lock

Module D treats the following theorem objects as frozen. Their internal proofs are not replayed and their conclusions are not broadened.

Package	Imported content	Module D use
Kernel/RPI	Kernel necessity; no repair; singleton physical interior; role coherence, closure, and minimality	Fixes the upstream target and excludes plural-interior or projection-backflow repairs
ATS	Anchor/tensor/skin trichotomy in a fixed domain	Fixes the eight work types but does not prove tensor assembly, skin maximality, candidate completeness, or fibre uniqueness
Structural Unification	One interior, paired projections, no backflow, joint-ledger and compatibility disciplines	Fixes the certificate incidence types
Module A	Source-bearing subtype, attachment, distributional source representation, Ward closure, finite composite closure	Supplies source generators, witness maps, and source-overlap data
Module B	Operator/source attachment, common Poincaré–Noether charge, active/inertial readouts, calibration	Supplies operator, charge, readout, and overlap generators
Module C	Independently formed upstream continuation, lawful metric orbit, full-ray quantum process, record/compatibility descent, directed exchange, cube	Supplies history, projected-process, boundary, incidence, and cube generators
Record theorem	Final measurement-record quotient for the fixed reuse profile	Supplies the record generator and boundary target point
Compatibility theorem	Final joint-witness quotient and compatibility-not-transport classification	Supplies the compatibility generator and boundary target point
Role-Occupancy Closure	Necessity and sufficiency of one compatible assignment orbit under a role-realization-complete certificate	Invoked only after all seven completeness coordinates are proved

Source-lock discipline. The exact imported PDFs, hashes, accepted review verdicts, tensor generator IDs, overlap IDs, and capstone contract are listed in `SOURCE_LOCK.md` and the ledgers in `ledgers/`. They fix the antecedent physical vertices and the tensor generator/overlap diagram before Module D candidate inspection. They do not insert the tensor colimit or skin kernel as antecedent conclusions; those are constructed in section 5.

3 The antecedent assembly package

Definition 3.1 (Frozen antecedent assembly data). Let

$$\Theta_D^{ABC,-} = (B_{\text{anc}}, B_{\text{GR}}, B_{\text{QM}}, B_{\text{bdry}}, B_{\text{trans}}, B_{\text{match}}, \mathbb{G}_D, (p_k)_k; (b_e)_{e \in E_D^{\text{req},-}})$$

be the exact source-locked package formed from accepted Modules A–C and the FEP-1.0 target. Its six physical vertices are:

1. B_{anc} : the target, domain, regularity, boundary, clock, source, scale, operator-domain, closed/open, failure, and report anchors;
2. B_{GR} : the lawful metric/source orbit and fixed gauge-reduced Einstein–matter development data;
3. B_{QM} : the full-ray projective process, self-adjoint generator/domain, potential/source envelope, initial ray, and clock-conditioned Schrödinger representation;
4. B_{bdry} : the final record and compatibility quotients with their descended processes;
5. B_{trans} : the fixed-clock RPI history quotient, standing/failure structure, common incidence, and strict/open continuation data;
6. B_{match} : the common-charge comparison, four-role intertwiners, directed exchange cells, role restrictions, and sixteen-vertex coherence package.

The package also contains the exact tensor generator/overlap diagram $\mathbb{G}_D$, its accepted probe cocone $(p_k)_k$, and all required incidence arrows that do not depend on the yet-to-be-constructed tensor and skin vertices. The superscript “ $-$ ” records that this is antecedent data, not yet the complete eight-vertex target diagram.

Proposition 3.2 (Antecedent package precedes Module D residue). *Every object and arrow in $\Theta_D^{ABC,-}$ is determined by the frozen target and accepted packages before Module D defines a tensor colimit, a skin congruence, a candidate comparison object, an eligible residue, a live residue, or an assignment.*

Proof. The six physical vertices and non-tensor/non-skin incidences are exact accepted theorem outputs. The tensor generator and overlap objects are the source-locked component interfaces and common subinterfaces listed in section K. Their accepted probe maps are also source locked. No tensor colimit, global joint probe, skin kernel, candidate, or residue is used to form this package. The missing tensor and skin vertices are constructed functorially from it in the next section. $\square$

Remark 3.3 (Base diagram notation after construction). After theorems 5.17 and 5.24, the notation $\mathbb{B}_D^{ABC}$ denotes the complete eight-vertex diagram obtained by adjoining $B_{\text{ten}} = L_D^0$ and $B_{\text{skin}} = S_D^0$ to $\Theta_D^{ABC,-}$. This later notation does not retroactively place either conclusion in the antecedent package.

4 The exact eight-sorted signature and antecedent target data

The correction distinguishes four objects that must not be conflated:

1. the *generic interface moduli*, in which many GR, quantum, boundary, transport, matching, tensor-diagram, and probe objects may exist;
2. the *source-locked antecedent data* supplied by the accepted Modules A–C certificate before any Module D candidate is inspected;

3. the tensor and skin target vertices constructed functorially from those antecedent data;
4. the raw presentations and partial responses generated only after the complete eight-vertex target diagram has been formed.

Module D proves uniqueness only in presentation fibres over the resulting fixed target points. It does not collapse the generic moduli.

Definition 4.1 (Forced certificate role set). The forced coordinate roles are

$$\mathcal{R}_D^{gg} = \{r_{\text{anc}}, r_{\text{ten}}, r_{\text{skin}}, r_{\text{GR}}, r_{\text{QM}}, r_{\text{bdry}}, r_{\text{trans}}, r_{\text{match}}\}.$$

They denote, respectively, the anchor envelope, tensor/operator/source ledger, lawful skin congruence, GR/source projection, projective-Hilbert quantum projection, boundary record/compatibility projection, upstream transport/continuation object, and matching/intertwining package.

Definition 4.2 (Generic interface moduli). For each role r , let M_r be the groupoid of mathematical objects satisfying the *generic* interface type of that role, with lawful role-preserving isomorphisms as arrows. The relevant data are:

1. M_{anc} : target, domain, regularity, boundary, clock, scale, operator-domain, and scope envelopes;
2. M_{ten} : finite source-typed standing-bearing generator diagrams together with lawful cocone presentations, not assumed colimiting;
3. M_{skin} : lawful joint probe functors together with candidate presentation congruences, not assumed maximal;
4. M_{GR} : lawful metric/source orbits and gauge-reduced Einstein–matter Cauchy data;
5. M_{QM} : full-ray projective processes with self-adjoint realization data;
6. M_{bdry} : record and compatibility quotient data with descended processes;
7. M_{trans} : fixed-clock RPI history data, seed, failure boundary, and incidence object;
8. M_{match} : charge comparisons, projection intertwiners, directed exchange, and cubical coherence.

These moduli are not asserted to be singleton. Different potentials, initial rays, source data, RPI seeds, generator diagrams, or lawful probe packages generally determine different objects in the corresponding moduli groupoid.

Definition 4.3 (Certificate indexing category). Let J_D^{gg} be the finite category whose objects are the eight roles and whose generating arrows are the exact cross-coordinate incidences listed in section 11. Composite arrows encode the corresponding composite ancestry, quotient, continuation, and matching relations.

Definition 4.4 (Source-locked antecedent package). The accepted capstone certificate supplies, before Module D candidate formation:

1. the anchor, GR/source, quantum, boundary, transport, and matching target points;
2. the exact accepted standing-bearing generator objects and overlap maps listed in section K;
3. the accepted complete probe and report maps on those generators;

4. the non-tensor/non-skin arrows of the required $E01$ – $E25$ incidence registry.

Denote this package by $\Theta_D^{ABC,-}$. It contains the *generator diagram* for the tensor vertex, not a tensor-colimit conclusion, and it contains the *probe data* from which the skin vertex will be formed, not a skin-maximality conclusion.

Proposition 4.5 (Antecedent independence from residue). *The role set, generic moduli, indexing category, and source-locked antecedent package are fixed before tensor-colimit construction, skin construction, candidate presentation formation, eligibility, live residue, automorphisms, or assignments.*

Proof. The role set and generic interfaces come from the prior certificate grammar. The six non-tensor/non-skin target points, the accepted generator objects, overlap maps, and probe maps are exact outputs of the frozen source-locked theorem packages. None is selected by inspecting Module D candidates. The tensor and skin vertices are constructed from these antecedent data in section 5; candidate formation begins only afterward. $\square$

Generic multiplicity is retained. A different bounded potential, constraint-satisfying initial-data set, initial ray, RPI seed, source attachment, tensor generator diagram, or probe functor may be legitimate in the relevant generic moduli. It is not silently called ineligible. It lies over a different target parameter and requires the corresponding reset certificate. Module D proves one lawful presentation class only in the fibres over the target points constructed from the exact accepted antecedent package.

5 Explicit tensor-colimit and dependent skin construction

This section discharges the constructional burden that cannot be placed in the target lock. The input is not a family of globally supported tensor diagrams. It is the source-locked *core* data supplied by RPI/AER and accepted Modules A–C, together with the exact physical incidence schema. The first task is therefore to construct lawful covariant completions of those partial interfaces. Only after that support-domain completion is proved do we form the tensor colimit, the induced complete probe, and the dependent skin kernel.

5.1 The lawful tensor-interface category

Definition 5.1 (Finite tensor-incidence schema). Let $\mathcal{S}_D$ be the finite category generated by the sixteen tensor-ledger sorts and the twenty-five source-to-target incidence arrows fixed in section K. Its objects include the upstream response, source attachment, Ward/regularity, composite exchange, normalized-ray source, operator EMT, common charge, active/inertial readout, RPI history, metric process, quantum process, record, compatibility, common incidence/cube, failure/report, and overlap-registry sorts. The only nonidentity generating arrows are $A01$ – $A25$; identities and composites are those of the generated finite acyclic category.

Write

$$\widehat{\mathcal{S}}_D^{\text{cov}} := \mathbf{Set}^{\mathcal{S}_D}$$

for the category of covariant set-valued diagrams on the physical incidence schema. Whenever the registry contains

$$u : s' \longrightarrow s,$$

a lawful ledger diagram carries a map in the same direction,

$$X(u) : X(s') \longrightarrow X(s).$$

Thus $A01 : S02 \rightarrow S01$ is represented by forgetting an attached source to its upstream response; no canonical attachment of an arbitrary response is introduced.

Definition 5.2 (Frozen witness/report covariant diagram). Let $\mathcal{W}_D \in \widehat{\mathcal{S}}_D^{\text{cov}}$ be the source-locked covariant diagram of accepted witness values, failure statuses, and report coordinates supplied by RPI/AER and accepted Modules A–C. Define

$$\mathbf{TenInt}_D := \widehat{\mathcal{S}}_D^{\text{cov}} / \mathcal{W}_D.$$

An object is a typed covariant ledger diagram $X : \mathcal{S}_D \rightarrow \mathbf{Set}$ with a lawful witness/report map $w_X : X \rightarrow \mathcal{W}_D$. A morphism is a natural transformation commuting with the witness/report maps.

Proposition 5.3 (Incidence-variance lock). *For every schema arrow $u : s' \rightarrow s$, every object (X, w_X) of $\mathbf{TenInt}_D$ carries the same-direction incidence map $X(u) : X(s') \rightarrow X(s)$. The complete A01–A25 registry is therefore typed exactly as the accepted physical interfaces listed in section K.*

Proof. This is the defining variance of an object of $\mathbf{Set}^{\mathcal{S}_D}$. Naturality of w_X gives $w_{X,s}X(u) = \mathcal{W}_D(u)w_{X,s'}$, so the physical incidence and its witness/report coordinate are transported in the same registered direction. $\square$

Lemma 5.4 (Cocompleteness of the tensor-interface category). *$\mathbf{TenInt}_D$ has all small colimits, and the forgetful functor to $\widehat{\mathcal{S}}_D^{\text{cov}}$ creates them.*

Proof. The functor category $\mathbf{Set}^{\mathcal{S}_D}$ is cocomplete and computes colimits objectwise. A colimit of a diagram in the slice carries the unique induced map to $\mathcal{W}_D$ obtained from the cocone of structure maps. Hence the slice is cocomplete and its colimits are created by the forgetful functor. $\square$

5.2 Core interfaces and covariant support completion

The source packages do not claim nonempty data at every schema sort. Each accepted generator first supplies a source-locked core diagram on the sorts where that theorem package natively lives. The Cycle-4 review correctly observed that extension by the empty set is not covariant whenever an arrow leaves a nonempty core sort. The lawful completion is instead the left Kan extension along the source-locked primary incidence inclusion. A sort is listed in a primary support only when the accepted package supplies a certified nonempty native interface there; the completion never inserts a dummy element.

Definition 5.5 (Core generator diagram). For each $k \in \{1, \dots, 15\}$, let $\mathcal{C}_k \subseteq \mathcal{S}_D$ be the source-locked primary incidence subcategory generated by the certified nonempty primary support recorded in `tensor_generators.csv` and the registered arrows whose source and target both lie in that core. Let

$$j_k : \mathcal{C}_k \hookrightarrow \mathcal{S}_D$$

be the faithful inclusion. The accepted theorem package supplies an object

$$(G_k^\circ, w_k^\circ) \in \mathbf{Set}^{\mathcal{C}_k} / j_k^* \mathcal{W}_D,$$

including every accepted incidence map whose source and target are both core sorts.

Definition 5.6 (Cosieve-completed generator). Define the completed generator by

$$\overline{G}_k := \text{Lan}_{j_k} G_k^\circ.$$

For $t \in \mathcal{S}_D$, this set is explicitly

$$\overline{G}_k(t) = \text{colim}_{(s,p) \in (j_k \downarrow t)} G_k^\circ(s).$$

Equivalently, its elements are classes $[s, p, x]$ with $s \in \mathcal{C}_k$, $p : s \rightarrow t$ in $\mathcal{S}_D$, and $x \in G_k^\circ(s)$, modulo the standard comma-category relations. For $u : t \rightarrow t'$ define

$$\overline{G}_k(u)[s, p, x] := [s, u \circ p, x].$$

The core witness map determines, by the Kan-extension adjunction, a unique natural map

$$\overline{w}_k : \overline{G}_k \longrightarrow \mathcal{W}_D.$$

Theorem 5.7 (Generator support-domain completion). *For every accepted generator:*

1. $(\overline{G}_k, \overline{w}_k)$ is an object of TenInt_D ;
2. the unit $\eta_k : G_k^\circ \rightarrow j_k^* \overline{G}_k$ is the canonical core comparison, and every identification in its image is generated only by the accepted primary incidence maps;
3. the nonempty support of $\overline{G}_k$ is the forward closure of the core support and is therefore a cosieve;
4. every schema arrow with source in the completed support has the explicitly constructed postcomposition map above.

Proof. The objectwise comma-category formula is the standard construction of Lan_{j_k} . Postcomposition defines a covariant functor because identities and composition are preserved on representatives. The unit sends a primary element to its identity-path class. The comma-category relation can identify such classes only through the accepted primary incidence maps in $\mathcal{C}_k$; no external placeholder or undeclared relation enters. If no path from a core sort reaches t , the comma category is empty and $\overline{G}_k(t) = \emptyset$; if a path exists from a nonempty core interface, the colimit is nonempty. Thus support is exactly the reachability closure. If t is supported and $u : t \rightarrow t'$, postcomposition sends a representing path to a path ending at t' , so t' is supported and the required map exists. Finally, the natural transformation $w_k^\circ : G_k^\circ \rightarrow j_k^* \mathcal{W}_D$ corresponds under adjunction to the unique $\overline{w}_k$, making the completion an object of the slice. $\square$

The machine-readable registry `tensor_generator_incidence_maps.csv` lists all 222 required generator-arrow maps. The completed supports in `tensor_generators.csv` are the exact forward closures of their source-locked core supports.

Definition 5.8 (Core overlap relation). For each $e \in \{1, \dots, 24\}$, let $\mathcal{C}_e \subseteq \mathcal{S}_D$ be the source-locked primary incidence subcategory generated by the core shared sorts. The accepted packages supply a common-interface object

$$O_e^\circ \in \mathbf{Set}^{\mathcal{C}_e} / j_e^* \mathcal{W}_D$$

and two lawful core maps to the restrictions of the adjacent accepted generators.

Definition 5.9 (Completed overlap relation). Let

$$\tilde{O}_e := \text{Lan}_{j_e} O_e^\circ.$$

Adjunction extends the two core maps to natural transformations

$$\tilde{i}_e : \tilde{O}_e \rightarrow \overline{G}_{\lambda(e)}, \quad \tilde{j}_e : \tilde{O}_e \rightarrow \overline{G}_{\rho(e)}.$$

Define $\overline{O}_e$ to be the image in the slice of the joint map

$$(\tilde{i}_e, \tilde{j}_e) : \tilde{O}_e \longrightarrow \overline{G}_{\lambda(e)} \times_{\mathcal{W}_D} \overline{G}_{\rho(e)}.$$

Write

$$i_e : \overline{O}_e \rightarrow \overline{G}_{\lambda(e)}, \quad j_e : \overline{O}_e \rightarrow \overline{G}_{\rho(e)}$$

for the two natural projections. Their joint map to the fibre product is monic; individual projections need not be monic after lawful incidence forgetting.

Theorem 5.10 (Overlap support-domain completion). *Every $\overline{O}_e$ is an object of TenInt_D whose support is the forward closure of the core shared sorts. Every schema arrow with supported source has a registered postcomposition map, the two overlap legs are natural over $\mathcal{W}_D$, and the completed relation contains only incidence-path images of the accepted core overlap.*

Proof. The left Kan extension supplies actual sets and postcomposition maps on the full forward closure. Images and fibre products in a set-valued functor category and its slice are computed objectwise, so the joint image is a covariant subdiagram of the product over $\mathcal{W}_D$. Its projections are natural and lawful. The unit maps the accepted core relation into its incidence completion, and the joint-image construction imposes no relation beyond the two accepted completed legs. A nonempty core element produces a nonempty completed pair along every reachable path; hence the support is the forward closure and is a cosieve. $\square$

The machine-readable registry `tensor_overlap_incidence_maps.csv` lists all 208 required overlap-arrow maps. The 975-row `tensor_support_closure_audit.csv` verifies, for every generator or overlap and every schema arrow, that a supported source has a supported target and a registered map.

5.3 The source-locked completed generator/overlap diagram

Definition 5.11 (Generator-overlap indexing category). Let $\mathbf{K}_D^{\text{ten}}$ be the finite bipartite category with generator objects $G01, \dots, G15$, overlap objects $O01, \dots, O24$, and for each overlap exactly two nonidentity arrows to its left and right generators, as fixed by the exact overlap registry. Identities and those arrows are the only generating maps.

Definition 5.12 (Accepted tensor-generator diagram). Define

$$\mathbb{G}_D : \mathbf{K}_D^{\text{ten}} \longrightarrow \text{TenInt}_D$$

by sending Gk to the completed generator $(\overline{G}_k, \overline{w}_k)$, Oe to the completed overlap relation $\overline{O}_e$, and the two overlap arrows to i_e and j_e . No colimit object is included in this antecedent diagram.

Theorem 5.13 (Covariant realization of the source-locked diagram). *The functor $\mathbb{G}_D$ is a type-correct source-locked diagram in $\mathbf{TenInt}_D$. All generator and overlap supports are forward-closed cosieves; every required incidence map is constructed; and the accepted core interfaces are recovered without alteration.*

Proof. Apply theorems 5.7 and 5.10. The support-completion certificate records fifteen primary diagrams, fifteen completed generator diagrams, twenty-four primary overlap relations, twenty-four completed joint-image overlap relations, 222 generator incidence maps, 208 overlap incidence maps, 309 shortest primary-to-target witness paths, and 314 completed overlap-embedding components. Every completed map is obtained either from an accepted core incidence or by the explicit postcomposition formula. Hence no map to an unsupported empty target is required. $\square$

Remark 5.14 (What is imported and what is new). The accepted modules supply the core interfaces, core incidence maps, core witness maps, and core overlap relations. Module D constructs their lawful covariant completion and then proves the total tensor-ledger universal property. The completion adds only incidence images forced by the frozen schema; it adds no new independent standing-bearing generator.

5.4 Objectwise construction of the tensor ledger

For notational economy, write $G_k := \overline{G}_k$ and $O_e := \overline{O}_e$ from this point onward. For a schema sort s , define an equivalence relation $\approx_s$ on

$$\prod_{k=1}^{15} G_k(s)$$

as the smallest equivalence relation generated by

$$(\lambda(e), i_{e,s}(x)) \approx_s (\rho(e), j_{e,s}(x))$$

for every $x \in O_e(s)$.

Definition 5.15 (Constructed tensor ledger). Define

$$L_D^0(s) := \left(\prod_{k=1}^{15} G_k(s) \right) / \approx_s .$$

For $u : s' \rightarrow s$ and $x \in G_k(s')$, define

$$L_D^0(u)[k, x] := [k, G_k(u)x].$$

The witness/report map is

$$w_{L,s}[k, x] := w_{G_k,s}(x).$$

Lemma 5.16 (Well-defined lawful covariant diagram). *The formulas above are independent of representatives, define a covariant functor $L_D^0 : \mathcal{S}_D \rightarrow \mathbf{Set}$, and make (L_D^0, w_L) an object of $\mathbf{TenInt}_D$.*

Proof. It suffices to check one generating overlap relation. Naturality of i_e and j_e sends a related pair at s' to the pair determined by $O_e(u)x$ at s . Therefore the equivalence is respected by every incidence map. Functoriality follows from that of the completed generators. Since both overlap legs lie over $\mathcal{W}_D$, witness values agree on every generating relation, so w_L is well defined and natural. $\square$

For each generator define

$$\iota_k : G_k \longrightarrow L_D^0, \quad \iota_{k,s}(x) = [k, x].$$

The two composites out of every completed overlap agree, so $(L_D^0, (\iota_k))$ is a cocone over $\mathbb{G}_D$.

Theorem 5.17 (Explicit tensor-colimit theorem). *The cocone $(L_D^0, (\iota_k))$ is a colimit of $\mathbb{G}_D$ in $\mathbf{TenInt}_D$.*

Proof. Let (Y, w_Y) be lawful and let $(\ell_k : G_k \rightarrow Y)_k$ be a cocone. Define

$$u_s([k, x]) = \ell_{k,s}(x).$$

The cocone equations identify the two images of every completed overlap element, so u_s is well defined on the least generated relation. Naturality and compatibility with $\mathcal{W}_D$ follow from those of the ℓ_k . Every class has a generator representative, so any mediator agreeing with the cocone maps agrees with u on every element. Thus the mediator exists and is unique. $\square$

Theorem 5.18 (Exact-overlap and no-extra-load theorem). *The constructed ledger has the following properties.*

1. *Every ledger element is represented by an element of an accepted completed generator, hence ultimately by accepted core data and a lawful incidence path.*
2. *Two generator elements are identified exactly when connected by a finite zigzag generated by the completed declared overlap relations.*
3. *No standing-bearing core generator, incidence image, or overlap relation outside the source-locked completion occurs in L_D^0 .*
4. *A cocone with independent extra load or an extra identification not generated by the declared completed overlaps is not a colimit presentation.*

Proof. Items 1–3 follow from the tagged coproduct, the explicit Kan completions, and the least generated equivalence relation. For item 4, the unique map from L_D^0 to a candidate cocone is non-surjective if independent extra load is present and non-injective if an undeclared identification is imposed. If the candidate were also colimiting, colimit uniqueness would force that map to be an isomorphism, a contradiction. $\square$

Definition 5.19 (Constructed tensor target point). The tensor target point is

$$\theta_{\text{ten}} := [L_D^0]_{\cong_{\mathbf{TenInt}_D}}.$$

It is a conclusion of theorem 5.17, not an antecedent assumption.

5.5 Induced joint probe and the skin vertex

Let $(\mathcal{V}_D, w_V)$ be the source-locked probe object of $\mathbf{TenInt}_D$. Each accepted core probe

$$p_k^\circ : G_k^\circ \rightarrow j_k^* \mathcal{V}_D$$

extends uniquely by the Kan adjunction to a slice morphism

$$p_k : G_k \rightarrow \mathcal{V}_D.$$

Core overlap compatibility extends to the completed overlap relation because the two composites agree before Kan extension and hence after adjunction; restriction to the joint image preserves that equality. Therefore $(p_k)_k$ is a cocone over the completed diagram.

Corollary 5.20 (Induced tensor probe). *There exists a unique slice morphism*

$$p_{\text{ten}} : (L_D^0, w_L) \longrightarrow (\mathcal{V}_D, w_V)$$

with $p_{\text{ten}} \iota_k = p_k$ for every generator.

Proof. Apply theorem 5.17 to the completed probe cocone. $\square$

Definition 5.21 (Frozen joint target-evaluation functor). Let

$$\text{Probe}_{\Theta_D^{ABC}} : \text{Pres}_D^- \longrightarrow \text{Witness}_D$$

be the product of the frozen anchor probes, the induced tensor probe, and the accepted GR/source, quantum, boundary, transport, matching, failure, and report probes. Here Pres_D^- is formed from the six source-locked non-tensor/non-skin vertices together with the constructed tensor vertex and all antecedent incidence maps.

Theorem 5.22 (Joint-probe completeness after tensor assembly). *Within the exact frozen target envelope, every standing-bearing presentation difference changes at least one component of $\text{Probe}_{\Theta_D^{ABC}}$. A lawful difference invisible to all components is presentation skin.*

Proof. Accepted Modules A–C supply complete or final core probe families in their exact envelopes. The Kan extensions evaluate every forced incidence image because each completed element is represented by core data and a path, and the completed probe is obtained by postcomposition along that path. By theorem 5.18, the tensor vertex contains every accepted core generator, every forced incidence image, and every completed overlap relation, with no extra load or identification. Thus a tensor difference changes a core probe, an incidence-image probe, an overlap relation, or a witness/report value; otherwise it is a lawful relabelling of the same completed colimit. The other role probes are complete by their accepted interfaces. ATS then classifies a same-target difference changing no anchor or standing-bearing probe as skin. $\square$

Definition 5.23 (Constructed skin target point). Let

$$S_D^0 := \text{Eq} \left(\text{Probe}_{\Theta_D^{ABC}} \circ \text{pr}_1, \text{Probe}_{\Theta_D^{ABC}} \circ \text{pr}_2 \right) \rightrightarrows \text{Pres}_D^-$$

be the kernel-pair groupoid of the joint target-evaluation functor, restricted to lawful same-target arrows. Define

$$\theta_{\text{skin}} := [S_D^0]_{\cong}.$$

Theorem 5.24 (Dependent skin construction and maximality). *S_D^0 is the unique maximal lawful same-target congruence through which every component of $\text{Probe}_{\Theta_D^{ABC}}$ factors.*

Proof. A kernel pair is a congruence and every congruence through which the functor factors is contained in it. Quotienting by the kernel pair gives the coarsest lawful quotient on which the joint probe is faithful. By theorem 5.22, no standing-bearing difference lies in the kernel. Therefore S_D^0 is sound and maximal, uniquely up to kernel-pair isomorphism. $\square$

5.6 Formation of the complete target diagram

Definition 5.25 (Frozen target-parameter diagram). Insert θ_{ten} and θ_{skin} into the six source-locked non-tensor/non-skin vertices. Add the tensor incidence arrows induced by the completed colimit injections and accepted projection maps, the skin factorization arrows induced by the kernel-pair quotient, and the remaining source-locked arrows of $E01$ – $E25$. The resulting functor

$$\Theta_D^{ABC} : J_D^{gq} \longrightarrow \text{Int}$$

is the complete frozen target-parameter diagram. Write $\theta_r = \Theta_D^{ABC}(r)$.

Theorem 5.26 (Full target-lock theorem). *The complete diagram Θ_D^{ABC} is constructed and fixed before Module D forms candidate presentations, eligibility edges, live residue, automorphisms, or assignments. Its tensor and skin vertices are uniquely determined by the source-locked core package and its lawful support completion, and the full diagram is unique up to lawful natural isomorphism.*

Proof. The six non-tensor/non-skin vertices, core generator/overlap interfaces, and core probe families are antecedent source-locked data. Generator and overlap completions are uniquely determined by their left Kan and image universal properties. The tensor vertex is then uniquely determined by the explicit colimit theorem, and the skin vertex by the dependent kernel-pair theorem. All occur before candidate formation. Lawfully isomorphic core presentations induce naturally isomorphic completions, colimits, probes, kernel pairs, and full target diagrams. $\square$

Proposition 5.27 (Role-set and target independence from residue). *The role set, generic interface moduli, indexing category, source-locked core package, support completions, constructed tensor and skin target points, and complete target diagram are fixed before residue inspection. None is inferred from the number or identity of candidate survivors.*

Proof. Combine proposition 4.5 and theorem 5.26. Candidate presentations are formed only after the target construction is complete. $\square$

6 Frozen standalone endpoint and nonclaims

Definition 6.1 (Module D endpoint). Module D closes when, for the exact pair $(\mathcal{C}_D^{gq}, T_D^{gq})$, it proves:

1. target-relative candidate completeness over $\mathbb{B}_D^{ABC}$;
2. presentation-quotient completeness in every role fibre;
3. role-set and eligibility completeness;
4. compatibility and failure completeness for the exact required edge registry;
5. one lawful target-eligible residue class in each of the eight fibres;
6. a live residue, independently formed from all compatible complete injections, equal to those eight classes;
7. a lawful representative transformation action;
8. exactly one compatible assignment orbit;
9. role-realization completeness and realization licensure by Role-Occupancy Closure.

Nonclaim 6.2 (No generic interface uniqueness). The theorem does not state that all objects of $\mathbf{l}_{GR}$, $\mathbf{l}_{QM}$, or the other broad interface categories are equivalent. Different sources, potentials, initial data, rays, seeds, or matching packages generally define inequivalent target instances.

Nonclaim 6.3 (No new physical dynamics). Module D does not construct a source, mass charge, clock, Einstein development, quantum flow, record process, interaction, or exchange channel.

Nonclaim 6.4 (No unrestricted quantum-gravity uniqueness). The result is limited to the exact frozen certificate and accepted Modules A–C. It does not classify richer field contents, global topologies, boundary conditions, high-curvature completions, or nonzero interacting quantum-gravity dynamics.

Nonclaim 6.5 (No cardinality-only proof). Eight roles and eight displayed base vertices do not suffice. The proof must establish target-equivalent eligibility, coherent incidence, live-residue formation, and one lawful orbit.

Nonclaim 6.6 (No generative use of Role-Occupancy Closure). The occupancy theorem is applied only after all candidate, quotient, role-set, eligibility, compatibility, live-residue, and failure data have been formed and proved complete.

Standalone review coordinates. Candidate formation, target-instance locking, presentation quotient, eight base-object interface theorems, exact edge completeness, live residue, representative orbit, software correspondence, and import authorization require separate verdicts.

Part II

Candidate Formation, Quotienting, Eligibility, and Failure

7 Certificate-generated candidates and target-fibred presentations

Role-Occupancy Closure does not ask whether every mathematical object in a broad physical envelope is uniquely determined. It asks whether every candidate response *generated by the fixed certificate* can occupy the target-forced role architecture in exactly one lawful way. The candidate class must therefore be formed from the operations of C_D^{ABC} , not from all objects that happen to share a Lorentzian or Hilbert-space envelope.

Definition 7.1 (Certificate response grammar). Let $\mathfrak{G}_D^{ABC}$ be the smallest typed response grammar containing the following constructors.

1. **Coordinate extraction:** for every role r , extract the accepted invariant vertex θ_r and any lawful presentation of it.
2. **Lawful redescription:** apply the coordinate, gauge, basis, phase, clock-presentation, source-representative, boundary, or notation isomorphisms licensed by the corresponding accepted interface.

3. **Interface restriction:** forget a declared subfamily of required operations or data, thereby producing a partial or incomplete coordinate response.
4. **Quotient and factorization:** form only the record, compatibility, source, projective, skin, and other quotient/factor objects already licensed by the accepted theorem packages.
5. **Incidence evaluation:** evaluate one of the generating arrows of J_D^{gg} , including ancestry, projection, continuation, exchange, matching, and report incidences.
6. **Failure/report formation:** attach the first failure or exact report status produced by the fixed audit grammar.
7. **Explicit reset:** replace a target parameter by a standing-bearing value outside Θ_D^{ABC} and mark the result as a new certificate instance rather than a same-target response.

No constructor invents a new physical source, potential, initial ray, RPI seed, interaction, boundary condition, or empirical selector.

Definition 7.2 (Generated candidate response set). The certificate-generated response set is

$$\text{Cand}_D^{ABC} := \{ \llbracket t \rrbracket : t \text{ is a well-typed term of } \mathfrak{G}_D^{ABC} \text{ not marked reset} \}.$$

The broad admissible candidate set used by Role-Occupancy Closure is the presentation-typed part of Cand_D^{ABC} surviving the upstream admissibility tests. Reset terms are recorded separately and never enter the same-target residue.

Definition 7.3 (Coordinate presentation fibration). For each role r , let

$$\pi_r : \mathcal{P}_r \longrightarrow \mathbf{M}_r$$

be the groupoid fibration whose objects are lawful interface presentations and whose base point is the standing-bearing invariant object represented by that presentation. The target fibre is

$$\mathcal{P}_r(\theta_r) := \{ P \in \mathcal{P}_r : \pi_r(P) \cong \theta_r \}.$$

A candidate presentation may lie elsewhere in $\mathcal{P}_r$, but it is then a presentation of a different target parameter, not a target-relative occupant of C_D^{ABC} .

Definition 7.4 (Generated coordinate candidate). A generated coordinate candidate of role type r is a pair

$$(P, \text{prov}_P)$$

where $P \in \mathcal{P}_r$ and prov_P is a derivation in $\mathfrak{G}_D^{ABC}$ showing how P was obtained from the fixed certificate. The provenance records whether the term is a complete target presentation, a lawful duplicate, a restriction/incomplete response, an incidence response, a failure, or a reset.

Definition 7.5 (Generic interface burden). For $b \in \mathbf{M}_r$, write $\text{lfc}_r(P; b)$ when P performs the generic role work over the base parameter b :

1. lfc_{anc} : P is a lawful presentation of the anchor tuple b ;
2. lfc_{ten} : P is a colimit cocone for the typed standing-bearing generator diagram contained in b ;
3. lfc_{skin} : P is the kernel congruence of the complete joint probe functor contained in b ;

4. Ifc_{GR} : P is a lawful presentation of the gauge-reduced Einstein–matter development determined by the source, gauge, regularity, initial, boundary, and exchange data in b ;
5. Ifc_{QM} : P is a faithful representative of the full-ray projective process, generator/domain, spatial/potential, initial-ray, and boundary data in b ;
6. Ifc_{bdry} : P is the product of the final record and compatibility quotients determined by the reuse, witness, and process data in b ;
7. $\text{Ifc}_{\text{trans}}$: P is the fixed-clock RPI Moore-history and directed-incidence object determined by the chart, clock, seed, standing, failure, and incidence data in b ;
8. $\text{Ifc}_{\text{match}}$: P is the comparison/intertwining cone determined by the source and target functors, common-charge map, incidence restrictions, exchange cells, and report data in b .

These predicates are uniform in b . They do not name the canonical presentations P_r^0 .

Definition 7.6 (Interface eligibility and target fidelity). For a generated candidate (P, prov_P) of sort r , define:

$$\text{IElig}_D(P, r) \iff \text{prov}_P \text{ is non-reset and complete} \wedge \text{Ifc}_r(P; \pi_r(P)) \wedge \text{NoHiddenLoad}_r(P).$$

Define target fidelity by

$$\text{TFid}_D(P, r) \iff \pi_r(P) \cong \theta_r$$

through a lawful invariant-interface isomorphism compatible with the incident arrows of Θ_D^{ABC} .

Definition 7.7 (Canonical presentations). Choose one accepted presentation $P_r^0 \in \mathcal{P}_r(\theta_r)$ for each role. Its lawful residue is denoted

$$\rho_r^0 = [P_r^0]_{\sim_r}.$$

The choice is a display convention. The target point θ_r and the eligibility predicates were fixed before this representative was selected.

Where exact values live. Source values, initial data, potential ledgers, initial rays, continuation seeds, and matching maps are neither smuggled into the generic role burden nor discarded. They are coordinates of the base parameter $b = \pi_r(P)$. Module D first allows the generic moduli to vary, then restricts to the fibre over the independently frozen point θ_r . The local theorems below are uniform fibre-finality theorems for every fixed b in the accepted domain.

8 Candidate completeness by generator closure

Definition 8.1 (Target-relevant certificate response). A target-relevant response of C_D^{ABC} is a possible occupant, branch, restriction, transport, correction, boundary readout, incidence readout, or failure report obtained by a finite licensed use of the fixed certificate operations. It is not an arbitrary object in one of the generic interface moduli.

Lemma 8.2 (Term normal form). *Every well-typed term of $\mathfrak{G}_D^{ABC}$ has a unique first outer constructor among: coordinate extraction/lawful presentation, restriction, quotient/factorization, incidence evaluation, failure/report formation, or reset.*

Proof. The response grammar is freely generated by the listed typed constructors subject only to the equations already imposed by lawful presentation functoriality, quotient universal properties, incidence composition, and report/failure idempotence. These equations do not change the first constructor class. Hence every term has a unique outer-constructor diagnosis. $\square$

Theorem 8.3 (Generated-candidate completeness). *Every target-relevant certificate response belongs to Cand_D^{ABC} or is explicitly marked as a reset. More precisely, every such response has exactly one primary disposition:*

1. a complete generated coordinate presentation in one of the eight target fibres;
2. a lawful presentation duplicate of such a coordinate;
3. an incomplete interface restriction;
4. a declared quotient/factorization output;
5. a cross-coordinate incidence response;
6. a first-failure or report object;
7. an explicit target reset.

Proof. Proceed by structural induction on the licensed construction of the response. The base responses are the eight coordinate extractions from Θ_D^{ABC} , hence are generated. Lawful redescription, restriction, quotient, incidence, and failure/report steps are constructors of $\mathfrak{G}_D^{ABC}$ and preserve membership in the generated class with the corresponding disposition. A standing-bearing operation not among these constructors changes one of the frozen certificate operations or adds a new operation; by the source lock it is a reset. The primary disposition is the first outer constructor from lemma 8.2. $\square$

Corollary 8.4 (Role-Occupancy candidate completeness). *The broad candidate class used in the imported Role-Occupancy theorem contains every admissible candidate response generated by C_D^{ABC} .*

Proof. By definition, Role-Occupancy candidates are possible occupants, branches, transports, corrections, sectors, readouts, or responses generated by the fixed certificate. Theorem 8.3 places every such response in the generated grammar, while resets are excluded from the same-target candidate set and recorded separately. $\square$

Theorem 8.5 (Eight-coordinate reduct theorem). *Every complete certificate presentation generated by C_D^{ABC} has a unique ordered tuple of eight coordinate presentations together with the incidence responses of J_D^{gg} :*

$$(P_{\text{anc}}, P_{\text{ten}}, P_{\text{skin}}, P_{\text{GR}}, P_{\text{QM}}, P_{\text{bdry}}, P_{\text{trans}}, P_{\text{match}}; \eta_e)_{e \in \text{Mor}_{\text{gen}}(J_D^{gg})}.$$

A generated response omitting one coordinate is an incomplete candidate, not a complete realization of the same certificate.

Proof. The complete-certificate constructor has the eight fixed coordinate slots and the generating incidence maps of J_D^{gg} . Coordinate extraction is deterministic, so the ordered reduct tuple is unique. Omitting a slot is the interface-restriction constructor and is therefore diagnosed as incomplete before eligibility or assignment is considered. $\square$

Theorem 8.6 (External-moduli separation). *Let $P \in \mathcal{P}_r$ satisfy the generic role interface but have*

$$\pi_r(P) \not\cong \theta_r.$$

Then P may define a legitimate object of the same broad physical envelope, but it is not a candidate response generated by the fixed certificate C_D^{ABC} unless an explicit reset constructor replaces θ_r by $\pi_r(P)$.

Proof. All non-reset coordinate-extraction terms begin at the fixed vertex θ_r , and lawful presentation constructors preserve its invariant isomorphism class. Restrictions and quotient/factorization constructors forget or factor data but do not replace the standing-bearing base point by an inequivalent one. Hence a complete object over a different base parameter cannot be produced without the reset constructor. The result is a new certificate instance, not a failed presentation of the old one. $\square$

Example 8.7 (Quantum moduli are not collapsed). Two real bounded potentials $V_0 \neq V_1$ on the same Hilbert carrier can both define valid objects of $\mathbf{M}_{\text{QM}}$. The fixed certificate contains one projective process and potential/source base point θ_{QM} . A presentation of the other potential is not declared mathematically false or “ineligible by preference”; it belongs to a reset certificate whose quantum base parameter has changed.

Remark 8.8 (Why this is not syntactic closure). The first release defined every same-envelope interface object to be a candidate and then attempted to force canonical values through eligibility. The repaired theorem uses the candidate class required by Role-Occupancy Closure: the responses actually generated by the fixed certificate. The finite grammar is justified by the exact accepted certificate operations, and alternatives in the generic moduli remain available to new certificates.

9 Lawful presentation groupoids and quotient completeness

The presentation quotient must collapse actual presentation duplication, not distinct physical values that happen to evade a selected probe family. The repaired quotient is therefore defined by lawful interface isomorphism over a fixed target parameter.

Definition 9.1 (Target presentation groupoid). For $b \in \mathbf{M}_r$, let $\mathcal{P}_r[b]$ be the full subgroupoid of $\mathcal{P}_r$ whose objects satisfy $\pi_r(P) \cong b$. An arrow

$$u : P \longrightarrow P'$$

is lawful over b iff it preserves the generic role operations and makes the invariant triangle commute up to the declared lawful natural isomorphism:

$$\pi_r(P) \xrightarrow{\pi_r(u)} \pi_r(P') \longrightarrow b.$$

Definition 9.2 (Presentation equivalence). For complete generated candidates P, P' in the same role fibre, write

$$P \sim_r P'$$

iff they lie in the same connected component of the lawful target presentation groupoid $\mathcal{P}_r[\theta_r]$. For partial candidates, the same definition is applied to the common partial interface parameter and never identifies a missing operation with an inhabited one.

Lemma 9.3 (Equivalence and congruence). *For every role r , the relation $\sim_r$ is an equivalence relation and a congruence for every operation and incident map preserved by the target presentation groupoid.*

Proof. Connectedness in a groupoid is reflexive, symmetric, and transitive. Every arrow is an interface isomorphism and therefore respects all role operations and incident maps. Composition and inversion preserve the commuting target triangle, so the equivalence is a congruence. $\square$

Definition 9.4 (Presentation-quotiented residue). The fibre residue is

$$\text{Residue}_r = \text{Ob}(\mathcal{F}_r^{\text{gen}}) / \sim_r,$$

where $\mathcal{F}_r^{\text{gen}}$ is the role- r part of $\mathbf{Cand}_D^{ABC}$. The total residue is the typed disjoint union

$$\text{Residue}(C_D^{ABC}) = \bigsqcup_{r \in \mathcal{R}_D^{gg}} \text{Residue}_r.$$

Proposition 9.5 (Quotient soundness). *If $P \sim_r P'$, then P and P' have the same interface eligibility, target fidelity, incident compatibility, continuation, boundary, failure, and report status.*

Proof. A lawful interface isomorphism over θ_r preserves the generic role operations, the invariant base point, every incident arrow of Θ_D^{ABC} , and the exact failure/report data. All listed predicates are invariant under such arrows. $\square$

Proposition 9.6 (No standing-bearing collapse). *If two complete candidates represent nonisomorphic invariant base points or differ in an incident standing-bearing map not related by a lawful interface isomorphism, then they are not equivalent under $\sim_r$.*

Proof. An arrow in $\mathcal{P}_r[\theta_r]$ must induce an invariant isomorphism over the fixed base and preserve all incident maps. If either condition fails, no such arrow exists. The presentations therefore remain distinct or the second object lies outside the target fibre. $\square$

Definition 9.7 (Profile functor). For audit purposes, each role carries a standing-bearing profile functor

$$\text{Inv}_r : \mathcal{P}_r \longrightarrow \mathcal{I}_r.$$

The profile is not the definition of presentation equivalence. It is complete on a target fibre only when the corresponding local theorem proves that equality of profiles lifts to a lawful interface isomorphism.

Theorem 9.8 (Quotient completeness from fibre finality). *Suppose the local role theorem proves that the target presentation groupoid $\mathcal{P}_r[\theta_r]$ is nonempty and connected and that every incomplete or nonfaithful generated candidate is diagnosed before occupation. Then $\sim_r$ collapses all and only presentation-level duplication in the target-relative eligible subfamily.*

Proof. Connectedness identifies every complete target-faithful presentation through actual lawful interface isomorphisms. Soundness follows from proposition 9.5. A nonisomorphic standing-bearing difference is retained by proposition 9.6; an incomplete or nonfaithful candidate is not promoted into the eligible target fibre. Thus the quotient removes exactly presentation multiplicity. $\square$

Remark 9.9 (Probe profiles are evidence, not quotient authority). In the tensor and skin fibres, final probe and kernel theorems help prove connectedness or uniqueness of the invariant object. They do not authorize the quotient to identify candidates merely because a discretionary test family fails to separate them.

10 Interface eligibility, target fidelity, and sort separation

Definition 10.1 (Sort fingerprint). The sort fingerprint $\chi(P)$ is the tuple of interface operations carried by a presentation. The decisive structures are:

Sort	Distinguishing interface structure
anchor	target-fixing domain and envelope maps
tensor	universal cocone over complete standing-bearing generators
skin	final kernel congruence of a probe functor
GR/source	lawful Lorentzian source orbit and Einstein–matter development
QM	full-ray projective process and self-adjoint implementation
boundary	final record and compatibility quotient product
transport	fixed-clock RPI Moore-history and common-incidence object
matching	common-charge comparison, natural intertwiners, exchange cells, and cube

Lemma 10.2 (Sort disjointness). *No generated coordinate presentation can occupy two distinct forced roles without first being retyped as two different reducts of a larger certificate presentation.*

Proof. The fingerprints have different operation types and universal burdens. A kernel congruence is not a Cauchy development; a full-ray process is not a record quotient; a Moore-history category is not an intertwining cone. A larger object may induce several typed reducts, but those reducts are distinct candidate responses. Hence a residue has at most one role type. $\square$

Definition 10.3 (Interface-eligible residue). For a residue $x = [P]_{\sim_r}$ and role r , write

$$\text{IEligRes}(x, r)$$

iff P is generated by C_D^{ABC} , has sort fingerprint r , satisfies $\text{IElig}_D(P, r)$, and the condition is invariant under $\sim_r$.

Definition 10.4 (Target-relative eligibility). The Role-Occupancy eligibility relation is

$$\text{Elig}_D(x, r) \iff \text{IEligRes}(x, r) \wedge \text{TFid}_D(P, r) \wedge \text{IncType}_r(P),$$

where P is any representative of x . Target fidelity compares the candidate invariant with the abstract target point θ_r ; it does not compare the candidate with the chosen canonical presentation P_r^0 .

Proposition 10.5 (Well-definedness of target-relative eligibility). *The relation Elig_D is independent of the representative P and is formed without using the canonical assignment or the unique-orbit conclusion.*

Proof. Interface eligibility, the moduli projection π_r , target fidelity, and incident endpoint typing are preserved by lawful interface isomorphisms by proposition 9.5. The target point θ_r is fixed by theorem 5.26; the canonical presentations and assignments are selected only later. Hence the relation is well defined and acyclic. $\square$

Theorem 10.6 (Eligibility completeness). *The relation Elig_D contains every and only same-certificate role incidence generated by C_D^{ABC} for the frozen target.*

Proof. By corollary 8.4, every candidate response generated by the certificate occurs in the generated grammar. The sort fingerprint determines which generic role work it can perform. The interface predicate tests that work without fixing a value. Target fidelity tests whether its invariant lies over the independently frozen target point, and the incident clause tests its local endpoints in the target diagram. A generated candidate satisfying all clauses is a same-certificate possible occupant. Failure of the generic interface, target fidelity, or incidence means respectively incomplete role work, a different certificate parameter, or a locally malformed cross-coordinate endpoint. No likelihood, preference, canonical-name test, or later utility creates an edge. $\square$

Corollary 10.7 (Single-role eligibility). *Every eligible residue is eligible for exactly one forced role.*

Proof. Existence follows from eligibility. Uniqueness follows from sort disjointness. $\square$

Eligibility does not collapse the moduli. A quantum object with a different potential can satisfy the generic quantum interface and therefore be interface-eligible in its own certificate. It fails target fidelity for the present θ_{QM} and is transferred to a reset certificate; it is not excluded by defining “quantum” to mean the canonical occupant.

11 Complete cross-coordinate incidence diagram

Pointwise target-relative eligibility is not global compatibility. A candidate assignment must present the entire frozen diagram Θ_D^{ABC} , not merely eight isolated vertices.

Definition 11.1 (Generating incidence registry). The generating arrows of J_D^{gq} are the following twenty-five target-fixed incidences.

- E1.** anchor $\rightarrow$ tensor: common target and envelope;
- E2.** anchor $\rightarrow$ skin: the probe kernel is formed on the same target;
- E3.** anchor $\rightarrow$ GR: metric/source data use the frozen Lorentzian and Cauchy envelope;
- E4.** anchor $\rightarrow$ QM: the quantum object uses the frozen Hilbert/domain/clock envelope;
- E5.** anchor $\rightarrow$ boundary: record and compatibility quotients use the fixed boundary scopes;
- E6.** anchor $\rightarrow$ transport: the RPI history uses the fixed domain, clock class, and failure envelope;
- E7.** anchor $\rightarrow$ matching: all matching maps preserve the report and calibration envelope;
- E8.** tensor $\rightarrow$ skin: every standing-bearing generator factors through the final skin congruence;
- E9.** tensor $\rightarrow$ GR: the GR source is the accepted Module A source in the tensor ledger;
- E10.** tensor $\rightarrow$ QM: the operator/source and inertial data agree with the tensor ledger and Module B;
- E11.** tensor $\rightarrow$ boundary: record and compatibility witnesses are the tensor-ledger boundary witnesses;

- E12.** tensor $\rightarrow$ transport: history, failure, and incidence data agree with Module C;
- E13.** tensor $\rightarrow$ matching: common source/operator ancestry and charge data agree across Modules A–B;
- E14.** transport $\rightarrow$ GR: the metric process is the Module C projection of the upstream history;
- E15.** transport $\rightarrow$ QM: the quantum process is the projection of that same history;
- E16.** transport $\rightarrow$ boundary: record and compatibility processes descend from the same history;
- E17.** transport $\rightarrow$ matching: the intertwiners and cube use the same upstream history and incidence;
- E18.** GR $\rightarrow$ matching: the metric intertwiner and active readout use the fixed GR/source object;
- E19.** QM $\rightarrow$ matching: the quantum intertwiner and inertial readout use the fixed QM object;
- E20.** boundary $\rightarrow$ matching: boundary intertwiners use the final quotient objects;
- E21.** skin $\rightarrow$ GR: GR presentation changes factor through the lawful skin congruence;
- E22.** skin $\rightarrow$ QM: quantum basis/phase/domain-core presentation changes factor through skin;
- E23.** skin $\rightarrow$ boundary: record and witness relabelling within final classes factors through skin;
- E24.** skin $\rightarrow$ transport: lawful RPI and clock-presentation changes factor through skin;
- E25.** skin $\rightarrow$ matching: natural-isomorphism and cube relabellings factor through skin.

The report status is an invariant attached to every vertex and arrow rather than an additional role.

Definition 11.2 (Compatible target presentation). Let

$$\tau : \mathcal{R}_D^{gq} \hookrightarrow \text{Residue}_{T_D^{gq}}^{\text{elig}}$$

be pointwise eligible. Choose representatives P_r and lawful target-fidelity isomorphisms

$$\lambda_r : \pi_r(P_r) \xrightarrow{\cong} \theta_r.$$

Define $\text{Compat}_D(\tau) = 1$ iff for every generating arrow $e : r \rightarrow s$ the square

$$\begin{array}{ccc} \pi_r(P_r) & \xrightarrow{\eta_e^\tau} & \pi_s(P_s) \\ \lambda_r \downarrow & & \downarrow \lambda_s \\ \theta_r & \xrightarrow{\Theta_D^{ABC}(e)} & \theta_s \end{array}$$

commutes, or satisfies the exact directed exchange relation declared by $\Theta_D^{ABC}(e)$; every local failure predicate is false; and the assembled report is the intersection of the accepted scopes.

Theorem 11.3 (Incidence-registry completeness). *The twenty-five generating incidences contain every standing-bearing cross-coordinate relation produced by C_D^{ABC} . Every composite compatibility condition is the image of a path in $\mathcal{J}_D^{gq}$ and follows from the generating squares.*

Proof. The capstone interface has only the following cross-sort operations: common anchor typing; probe/skin factorization; source, operator, record, history, and charge ancestry; the four role projections of the common history; the four matching/intertwining endpoints; and lawful presentation factorization. The listed arrows separate each endpoint pair explicitly. Accepted Module C cubical coherence and ordinary functoriality impose the path relations. By the candidate-generation theorem, any additional standing-bearing cross-coordinate operation would be a new constructor and therefore a reset certificate. Hence the registry is complete for the fixed certificate. $\square$

Theorem 11.4 (Canonical compatibility). *The accepted Modules A–C presentations determine a pointwise eligible assignment whose comparison squares commute for all twenty-five generating arrows. Hence the canonical assignment satisfies $\text{Compat}_D = 1$.*

Proof. The target-fidelity maps are the accepted source-locked interface identifications. Anchor squares commute by the common frozen envelope. Tensor ancestry and source/operator squares are the accepted Modules A–B attachment and charge theorems. History-projection and boundary squares are the accepted Module C descent/intertwining theorems. The matching squares and directed exchange relations are the accepted common-charge and continuation-cube results. The skin squares commute by lawful quotient/naturality. Every source package includes its exact report boundary, so the report intersection is preserved. $\square$

Remark 11.5 (Compatibility is diagrammatic, not canonical-name matching). The assignment is compatible because it presents the independently frozen incidence diagram. The proof does not compare string identifiers or require local candidates to be named after the accepted presentations.

12 Ordered failure exhaustion and explicit certificate reset

Definition 12.1 (First-failure pipeline). Audit a proposed response or certificate in the following order:

1. candidate generation or explicit reset;
2. coordinate/sort formation;
3. generic interface eligibility;
4. target-parameter fidelity;
5. lawful presentation quotienting;
6. cross-coordinate diagram compatibility;
7. live-residue formation;
8. assignment coverage and lawful-orbit closure.

The earliest failed stage is the primary diagnosis. A later correction forms a new act or a reset certificate; it does not retroactively change the earlier status.

Theorem 12.2 (Failure-completeness theorem). *Every non-realizing response generated by C_D^{ABC} , or every proposed external replacement, has at least one of the following diagnoses:*

1. malformed or incomplete generated candidate;

2. *external moduli value requiring explicit target reset;*
3. *wrong sort or failure of the generic role interface;*
4. *failure of target fidelity to the frozen parameter θ_r ;*
5. *unlawful presentation overcounting or standing-bearing quotient collapse;*
6. *failure of one or more generating incidence squares or directed exchange relations;*
7. *pointwise eligible but globally compatibility-impossible residue;*
8. *excess or missing live residue, non-bijective coverage, or more than one lawful assignment orbit;*
9. *explicit scope reset.*

No additional same-certificate repair route exists.

Proof. Generated-candidate completeness gives the first and incidence/report cases and separates reset terms. The sort and interface stages test whether the object performs the role work. Target fidelity distinguishes a presentation of the fixed certificate parameter from a legitimate object of another certificate. The groupoid quotient separates lawful presentation multiplicity from standing-bearing difference. Incidence-registry completeness tests every cross-coordinate relation generated by the certificate. Live-residue and orbit formation test complete occupation and selector freedom. These are exactly the formation stages of the imported Role-Occupancy object, so no same-certificate route lies outside the pipeline. $\square$

Corollary 12.3 (No empirical or semantic repair). *Likelihood, empirical fit, elegance, explanatory preference, semantic intention, or later usefulness cannot create candidate generation, interface eligibility, target fidelity, compatibility, or orbit closure. Such data may enter only as a newly declared boundary record in a reset certificate.*

Typical dispositions. A different bounded quantum potential is a new target parameter, not a rejected presentation of the fixed one. A source tensor with no Module A attachment fails the generic GR/source interface. A downstream pullback used as the upstream history fails transport formation. A lawful gauge or basis change remains in the same presentation groupoid. An extra live residue appearing in another complete compatible injection fails closure even if all local interfaces are satisfied.

Part III

Eight Independent Fibre-Exhaustion Theorems

13 Anchor-fibre exhaustion

Definition 13.1 (Anchor moduli and target point). An anchor parameter is a tuple

$$b = (D, T, \mathcal{E}, \mathcal{R}, \mathcal{B}, \mathcal{K}, \mathcal{S}, \mathcal{O}, \mathcal{C}) \in \mathbf{M}_{\text{anc}}$$

recording domain, target, admissibility envelope, regularity, boundary, clock, scale/calibration, operator-domain, and closed/open scope data. The frozen point θ_{anc} is the tuple fixed by FEP-1.0 and the exact accepted source locks.

Theorem 13.2 (Uniform anchor-presentation finality). *For every fixed anchor parameter $b \in \mathbf{M}_{\text{anc}}$, the groupoid of complete lawful presentations satisfying $\text{lfc}_{\text{anc}}(-; b)$ is nonempty when b is represented and connected under lawful renaming. Hence its presentation quotient contains one class.*

Proof. An anchor presentation displays the components of b . Two such displays differ only by names, equivalent encodings, or lawful coordinate choices that leave each component of b unchanged. The componentwise identity on the invariant tuple lifts to a lawful presentation isomorphism. A substantive change gives a different parameter b' , not a second presentation over b . $\square$

Corollary 13.3 (Target-relative anchor finality). *The eligible anchor residue over the frozen target point is singleton:*

$$\text{Residue}_{\text{anc}}^{\text{elig}} = \{\rho_{\text{anc}}\}.$$

Proof. Apply theorem 13.2 to $b = \theta_{\text{anc}}$. The accepted presentation supplies nonemptiness; target fidelity excludes other base points without declaring them mathematically impossible. $\square$

Corollary 13.4 (Anchor rigidity). *Every certificate-lawful role-incidence automorphism fixes the anchor role and its live residue.*

14 Tensor/operator/source-fibre exhaustion

The tensor target point is no longer inserted by the source lock. It is the lawful isomorphism class of the explicitly constructed colimit L_D^0 from theorem 5.17. The generic tensor moduli still contain arbitrary lawful cocones and arbitrary generator diagrams; target fidelity is the additional requirement that a candidate present the colimit of the exact source-locked diagram.

Definition 14.1 (Generic tensor cocone presentation). For a fixed typed generator diagram

$$\mathbb{G}_b : \mathbf{K}_b \longrightarrow \mathbf{TenInt}_D,$$

a tensor presentation is a lawful cocone

$$(L, (\lambda_k : G_k \rightarrow L)_k).$$

It is *interface eligible* when it preserves the schema, witness/report map, and declared overlap cocone. It is *target faithful* when it has the colimit universal property for the fixed diagram $\mathbb{G}_b$.

This separates a correctly typed ledger display from a complete tensor ledger. A cocone may be lawfully typed while omitting a generator, adding independent load, or imposing an extra identification; such a cocone is not target faithful.

Theorem 14.2 (Uniform tensor-colimit existence). *For every finite covariantly source-typed diagram $\mathbb{G}_b$ in $\mathbf{TenInt}_D$, the objectwise tagged-coproduct quotient construction of definition 5.15 produces a colimit*

$$L_b^0 \cong \text{colim } \mathbb{G}_b.$$

Proof. The proof of theorem 5.17 is diagram-uniform. Replace the frozen generator and overlap IDs by the finite objects and arrows of $\mathbb{G}_b$. Since $\mathbf{TenInt}_D = \mathbf{Set}^{S_D}/\mathcal{W}_D$ is cocomplete and colimits are created objectwise, the same generated-equivalence construction and mediator formula prove the universal property. $\square$

Theorem 14.3 (Uniform tensor-colimit finality). *For every fixed diagram $\mathbb{G}_b$, any two target-faithful tensor presentations are uniquely lawfully isomorphic over the generator cocone. Hence the target-faithful presentation groupoid over b is nonempty and connected.*

Proof. Existence is theorem 14.2. Let (L, λ_k) and (L', λ'_k) be two colimit cocones. The universal property of L gives a unique lawful map $u : L \rightarrow L'$ with $u\lambda_k = \lambda'_k$, and the universal property of L' gives a unique lawful map $v : L' \rightarrow L$ with $v\lambda'_k = \lambda_k$. Both vu and id_L mediate the cocone (λ_k) , so uniqueness gives $vu = \text{id}_L$; similarly $uv = \text{id}_{L'}$. Thus u is the unique lawful isomorphism commuting with every generator injection. $\square$

Proposition 14.4 (Dropped, duplicated, extra, and overidentified ledgers fail target fidelity). *Let (L, λ_k) be an interface-eligible cocone for a fixed diagram.*

1. *If a source-locked generator has no image in L , the cocone is incomplete.*
2. *If L contains an independent standing-bearing element not generated by the λ_k , mediator uniqueness fails.*
3. *If two generator elements are identified without a finite zigzag of declared overlaps, the mediator to the explicit quotient colimit does not exist.*
4. *If one accepted generator is duplicated as two independent standing-bearing copies, the mediator from the explicit colimit is not surjective onto both copies and the cocone is not universal.*

Therefore none of these presentations is target faithful.

Proof. Item 1 violates the source-locked cocone interface. For items 2–4, let $c : L_b^0 \rightarrow L$ be the unique map induced from the explicit colimit by the candidate cocone. Extra independent load or an independently duplicated generator makes c non-surjective; an undeclared identification makes c non-injective. If the candidate cocone were also colimiting, colimit uniqueness would make c an isomorphism. Each failure is therefore detected by the universal property, not by comparison with a preferred display. $\square$

Definition 14.5 (Complete tensor probe on a target-faithful presentation). For a target-faithful presentation (L, λ_k) , let

$$\mathcal{P}_{\text{ten},b} : L \longrightarrow \mathcal{V}_b$$

be the unique map induced by the accepted compatible generator-probe cocone. Under the unique colimit comparison, it corresponds to the explicitly constructed probe on L_b^0 .

Lemma 14.6 (Probe-family adequacy on the frozen diagram). *On the source-locked diagram $\mathbb{G}_D$, the induced probe family evaluates every accepted generator sort, every declared overlap/incidence sort, and every failure/report coordinate. A lawful difference between target-faithful tensor presentations that is invisible to this family is presentation skin.*

Proof. The generator probes are complete in the exact accepted Module A–C envelopes. The tensor schema contains the incidence and overlap sorts, so an altered gluing map is itself probed. Every element of a target-faithful ledger lies in the image of a generator injection, and every relation is generated by a declared overlap. Hence a standing-bearing difference changes a generator probe, an incidence probe, or a report coordinate. If none changes, the unique colimit comparison identifies the two presentations, and ATS classifies the remaining difference as skin. $\square$

Corollary 14.7 (Target-relative tensor finality). *For the frozen source-locked diagram,*

$$\text{Residue}_{\text{ten}}^{\text{elig}} = \{\rho_{\text{ten}}\}.$$

Proof. The explicit object L_D^0 is a target-faithful presentation by theorem 5.17, so the fibre is nonempty. Uniform tensor-colimit finality gives one lawful presentation class over $\theta_{\text{ten}} = [L_D^0]$. $\square$

Remark 14.8 (Generic tensor moduli remain open). A different source, operator family, continuation, generator set, or overlap map produces a different diagram $\mathbb{G}_{b'}$ and may have its own colimit. Module D does not identify colimits of nonisomorphic diagrams. It proves existence and presentation finality for the exact source-locked diagram used by this certificate.

15 Skin-fibre exhaustion and maximality

The skin target is constructed only after the tensor ledger and its induced probes exist. This dependency is essential: a congruence cannot be called maximal standing-neutral skin until every source-locked tensor generator and overlap incidence has entered the joint target-evaluation functor.

Definition 15.1 (Joint probe functor at a base point). For a fixed complete certificate-parameter diagram b whose tensor vertex is a proved colimit presentation, let

$$\text{Probe}_b : \text{Pres}(b) \longrightarrow \text{Witness}(b)$$

be the product of the anchor, induced tensor, GR/source, quantum, boundary, transport, matching, failure, and report probes at b .

Definition 15.2 (Kernel congruence). The lawful skin congruence at b is the kernel pair of Probe_b within the lawful same-target presentation groupoid:

$$p \sim_{\text{skin},b} q \iff \text{Probe}_b(p) = \text{Probe}_b(q)$$

with a connecting lawful arrow preserving the base parameter and every incident map.

Theorem 15.3 (Uniform maximal-kernel theorem). *For every fixed complete b , $\sim_{\text{skin},b}$ is the unique maximal lawful congruence through which every component of Probe_b factors.*

Proof. The kernel pair of a fixed functor is a congruence, and every congruence through which the functor factors is contained in it. Conversely, quotienting by the kernel pair gives the coarsest lawful presentation quotient on which Probe_b is faithful. The restriction to lawful same-target arrows prevents the kernel from identifying anchor motion, target reset, or malformed cross-role maps merely because of a partial presentation. $\square$

Theorem 15.4 (Soundness of the frozen skin kernel). *For the complete target diagram constructed in section 5, the kernel pair of $\text{Probe}_{\Theta_D^{ABC}}$ contains no standing-bearing difference.*

Proof. Joint-probe completeness is theorem 5.22. The tensor component is complete because L_D^0 is the colimit of every accepted generator and declared overlap, and the remaining components are the accepted complete/final role probes. A pair in the kernel preserves all these values and all anchors. Therefore it changes no admitted standing-bearing use and is skin by ATS. $\square$

Corollary 15.5 (Target-relative skin finality). *For the frozen target diagram,*

$$\text{Residue}_{\text{skin}}^{\text{elig}} = \{\rho_{\text{skin}}\}.$$

Proof. The constructed skin object S_D^0 is the kernel pair of the fixed complete joint probe by definition 5.23. It is sound and maximal by theorems 5.24 and 15.4. Every eligible skin occupant must present this maximal kernel pair; kernel pairs of one functor are uniquely isomorphic over their base. Hence the target-relative skin fibre is nonempty and connected. $\square$

Remark 15.6 (No profile circularity). The probe functor is not selected to preserve a canonical tensor display. Its tensor component is induced from the independently source-locked generator probes through the proved colimit universal property, and its other components are accepted interface probes. The skin object is then the dependent kernel construction on that fixed functor, before candidate residue or orbit formation.

16 GR/source-fibre exhaustion

Definition 16.1 (GR/source base parameter). A GR/source base point

$$b_g \in M_{\text{GR}}$$

contains: a lawful metric/source orbit; an accepted Module A source attachment and Ward/exchange class; the minimal metric anchors; a gauge-reduced Einstein–matter system; regularity, initial, boundary,

and local-interval data; the Module C metric process and common-incidence restriction; and, where applicable, the Module B active scalar readout.

Definition 16.2 (GR/source presentation interface). A presentation G satisfies $\text{lfc}_{\text{GR}}(G; b_g)$ iff it is a lawful gauge/boundary/source presentation of the local Einstein–matter development determined by b_g , with the accepted source ancestry and directed-exchange status.

Theorem 16.3 (Uniform GR/source fibre finality). *For every fixed Cauchy-eligible base parameter b_g , the groupoid of presentations satisfying $\text{lfc}_{\text{GR}}(-; b_g)$ is connected. Hence its lawful presentation quotient contains one class.*

Proof. The base parameter fixes the source system, gauge reduction, regularity class, initial data, boundary data, local interval, and exchange current before any presentation is chosen. The Einstein normal-form theorem fixes the same-scope equation form. Accepted Module A fixes source ancestry and the distributional/smooth source class. Local gauge-reduced Cauchy uniqueness identifies the physical development for those fixed data. Module C identifies the projected RPI metric process with that development and fixes the directed exchange from the common incidence. Two presentations therefore differ only by lawful diffeomorphism, gauge, source representative, or boundary encoding and are connected in the target presentation groupoid. $\square$

Corollary 16.4 (Target-relative GR/source finality). *For the frozen base point θ_{GR} ,*

$$\text{Residue}_{\text{GR}}^{\text{elig}} = \{\rho_{\text{GR}}\}.$$

Proof. The accepted Modules A and C presentation supplies nonemptiness. Apply theorem 16.3 to the target base point. $\square$

Remark 16.5 (Cauchy uniqueness is parameter-fibred). Different source systems or constraint-satisfying initial data are different points of $\mathbf{M}_{\text{GR}}$. The theorem neither identifies them nor declares them ineligible in their own certificates. It proves uniqueness of lawful presentations and developments only in the fibre over one fixed data package.

17 Quantum-fibre exhaustion

Definition 17.1 (Quantum base parameter). A quantum base point

$$b_q \in \mathbf{M}_{\text{QM}}$$

contains: the projective-Hilbert carrier and full transition profile; the clock-conditioned strongly continuous projective process; a self-adjoint generator graph and domain; the scalar spatial and potential/source data; the initial ray and boundary certificate; the declared closed/open status; the accepted Module B inertial coordinate where applicable; and the Module C record/compatibility incidence.

Definition 17.2 (Quantum presentation interface). A presentation Q satisfies $\text{lfc}_{\text{QM}}(Q; b_q)$ iff it is a faithful projective-Hilbert representative of the fixed base process b_q , with the declared self-adjoint, spatial, source, initial-ray, boundary, and closure data.

Theorem 17.3 (Uniform quantum fibre finality). *For every fixed quantum base point b_q in the accepted full-ray envelope, the groupoid of presentations satisfying $\text{lfc}_{\text{QM}}(-; b_q)$ is connected under lawful unitary, basis, phase-baseline, graph-core, and equivalent-domain presentations.*

Proof. The full transition profile and projective process are fixed in b_q . Wigner's theorem gives unitary or antiunitary implementations of transition-preserving bijections; strong continuous time flow through the identity selects the unitary branch. The fixed projective process determines the unitary group up to the declared homogeneous phase baseline, and Stone's theorem determines its self-adjoint generator. The generator domain, potential/source ledger, spatial carrier, initial ray, boundary certificate, and closure status are parameters of b_q , not conclusions inferred from Wigner or Stone. The Schrödinger normal-form theorem fixes the minimal scalar spatial form within those parameters, and Module B fixes the inertial coefficient on its overlap. Hence every faithful presentation of the same b_q is lawfully unitarily equivalent. $\square$

Corollary 17.4 (Target-relative quantum finality). *For the frozen quantum parameter θ_{QM} ,*

$$\text{Residue}_{\text{QM}}^{\text{elig}} = \{\rho_{\text{QM}}\}.$$

Proof. The accepted Module C quantum presentation and Module B overlap supply a target-fibre inhabitant. Apply theorem 17.3. $\square$

Remark 17.5 (No all-Hamiltonian uniqueness). Different potentials, generator domains, initial rays, or projective processes define different base points of M_{QM} . Module D does not identify them. It proves one lawful presentation class over the fixed process and data generated by the accepted certificate.

18 Boundary record/compatibility-fibre exhaustion

Definition 18.1 (Boundary base parameter). A boundary base point $b_{\partial} \in \text{M}_{\text{bdry}}$ consists of a retained/prospective/deliberative/evidential reuse profile, a joint Bell/boundary witness profile, the no-transport status, declared boundary certificates, and the record and compatibility processes descending from a fixed upstream history.

Definition 18.2 (Boundary presentation interface). A boundary presentation satisfies $\text{lfc}_{\text{bdry}}(-; b_{\partial})$ iff it is the product of:

1. the final measurement-record quotient determined by the reuse profile of b_{∂} ;
2. the final compatibility quotient determined by the joint-witness and no-transport profile of b_{∂} ;
3. the exact descended record and compatibility processes specified in b_{∂} .

Lemma 18.3 (Uniform record quotient finality). *For every fixed reuse profile, any standing-bearing record presentation factors uniquely through the corresponding measurement-record quotient.*

Proof. Assignments with the same complete reuse profile are identified and those with different standing-bearing profiles are separated. The standard quotient universal property gives unique factorization for every admissible record map constant on the equivalence classes. $\square$

Lemma 18.4 (Uniform compatibility quotient finality). *For every fixed joint-witness/no-transport profile, any standing-bearing compatibility presentation factors uniquely through the corresponding final compatibility quotient.*

Proof. The compatibility theorem fixes the complete joint witness content and lawful local realizations while excluding hidden transport, source transfer, and repair. The quotient universal property gives unique factorization. $\square$

Theorem 18.5 (Uniform boundary-fibre finality). *For every fixed b_∂ , the groupoid of boundary presentations satisfying $\text{lfc}_{\text{bdry}}(-; b_\partial)$ is connected.*

Proof. The two quotient factors are unique up to unique isomorphism by lemmas 18.3 and 18.4. The descended processes are parameters of b_∂ and are exact quotient descents of the fixed upstream history. Product presentations therefore differ only by lawful labels, representative assignments, or equivalent quotient encodings. $\square$

Corollary 18.6 (Target-relative boundary finality). *For the frozen boundary point θ_{bdry} ,*

$$\text{Residue}_{\text{bdry}}^{\text{elig}} = \{\rho_{\text{bdry}}\}.$$

19 Transport/continuation-fibre exhaustion

Definition 19.1 (Transport base parameter). A transport base point $b_c \in \mathbf{M}_{\text{trans}}$ contains a regular local RPI quotient chart, a certified clock class and fixed representative, a globally valid standing-preserving seed, standing/failure boundary data, the common incidence object and role restrictions, and the declared strict/open status.

Definition 19.2 (Transport presentation interface). A presentation satisfies $\text{lfc}_{\text{trans}}(-; b_c)$ iff it is the fixed-clock Moore-history category generated by the seed in b_c , with no-repair prefix structure, lawful duration transport between clock presentations, common incidence, and the declared strict/open process data.

Theorem 19.3 (Uniform transport-fibre finality). *For every fixed transport parameter b_c satisfying the Module C regularity hypotheses, the groupoid of presentations satisfying $\text{lfc}_{\text{trans}}(-; b_c)$ is connected.*

Proof. The chart, clock representative, seed, standing submanifold, and failure boundary are fixed in b_c . Local flow uniqueness determines the semiflow. Moore segments are determined by that flow and their durations, and strict concatenation determines the history category. A lawful monotone clock change gives the accepted duration-transport equivalence between the corresponding presentation categories. Common incidence and open/closed status are parameters of b_c and are preserved. Hence all complete presentations are lawfully equivalent. $\square$

Corollary 19.4 (Target-relative transport finality). *For the accepted Module C parameter θ_{trans} ,*

$$\text{Residue}_{\text{trans}}^{\text{elig}} = \{\rho_{\text{trans}}\}.$$

Proof. Accepted Module C supplies a complete nonidentity inhabitant over the frozen base point. Apply theorem 19.3. $\square$

Remark 19.5 (No uniqueness of all seeds). A different smooth standing-preserving RPI seed is a different point of $\mathbf{M}_{\text{trans}}$. Local flow uniqueness never identifies different seeds. The repaired theorem is the uniform statement: for each fixed seed package, its lawful history presentations form one class.

20 Matching/intertwining-fibre exhaustion

Definition 20.1 (Matching base parameter). A matching base point $b_m \in \mathbf{M}_{\text{match}}$ contains:

1. the fixed Module B Poincaré–Noether/Casimir comparison and calibrated active/inertial readout maps;
2. the fixed Module C upstream continuation and the four target projection functors;
3. the closed-prediction and actual exchange-extended processes;
4. the common incidence and fixed role restrictions;
5. the source/target data of the directed exchange cells;
6. the sixteen-vertex cube and exact report intersection.

Definition 20.2 (Matching presentation interface). A presentation satisfies $\text{lfc}_{\text{match}}(-; b_m)$ iff it is a comparison cone whose intertwiners, directed exchange cells, cube edges, and report maps have exactly the source and target functors specified by b_m and satisfy the accepted naturality and cubical coherence laws.

Lemma 20.3 (Uniform closed-intertwiner uniqueness). *For fixed source and target functors in b_m , the closed metric, quantum, record, and compatibility intertwiners are unique.*

Proof. Metric intertwining is fixed by the accepted Cauchy development for the fixed source and data. Quantum intertwining is fixed by the accepted full-ray process and Stone implementation. Record and compatibility intertwiners are exact quotient descents. Hence another natural transformation with the same components and required action agrees on every object and morphism. $\square$

Lemma 20.4 (Uniform exchange-cell uniqueness). *For fixed common incidence and role restrictions in b_m , each directed exchange cell is unique.*

Proof. The source, target, enriched carrier action, and incidence increment are fixed. The exchange cell is the unique action morphism advancing the ledger by the declared restriction while leaving the closed base coordinate unchanged. $\square$

Lemma 20.5 (Uniform scalar-matching uniqueness). *For a fixed accepted Module B matching parameter and calibration, the active/inertial scalar comparison is unique.*

Proof. The attached operator EMT, translation charge, zero-transfer form factor, Casimir line, controlled Wigner-to-Bargmann contraction, and calibration are fixed in the base parameter. The two scalar readouts and their proportionality factor therefore have no remaining standing-bearing freedom. $\square$

Theorem 20.6 (Uniform matching-fibre finality). *For every fixed matching parameter b_m in the accepted envelope, the groupoid of presentations satisfying $\text{lfc}_{\text{match}}(-; b_m)$ is connected.*

Proof. Combine lemmas 20.3 to 20.5. The sixteen-vertex cube is generated functorially from the four fixed role cells; whiskered coordinatewise pasting fixes every face and monotone path. Remaining differences are lawful natural-isomorphism or labelling presentations. $\square$

Corollary 20.7 (Target-relative matching finality). *For the frozen matching point θ_{match} ,*

$$\text{Residue}_{\text{match}}^{\text{elig}} = \{\rho_{\text{match}}\}.$$

Part IV

Live Residue, Lawful Automorphisms, and Unique Occupancy

21 The target-diagram presentation assignment

Theorem 21.1 (Eligible residue theorem). *The target-relative eligible residue is exactly*

$$\text{Residue}_{T_D^{gq}}^{\text{elig}} = \{\rho_{\text{anc}}, \rho_{\text{ten}}, \rho_{\text{skin}}, \rho_{\text{GR}}, \rho_{\text{QM}}, \rho_{\text{bdry}}, \rho_{\text{trans}}, \rho_{\text{match}}\}.$$

Each element is eligible for exactly its corresponding role.

Proof. The eight uniform fibre-finality theorems, specialized to the independently fixed target points θ_r , give one lawful presentation class in each role fibre. Existence is supplied by the accepted Modules A–C coordinate presentations. Generated-candidate completeness and target fidelity exclude no external moduli object by fiat: such an object lies over another target point and belongs to a reset certificate. Sort disjointness gives exact single-role eligibility. $\square$

Definition 21.2 (Displayed target assignment). Choose one accepted presentation of each target vertex and define

$$\sigma_0 : \mathcal{R}_D^{gq} \longrightarrow \text{Residue}_{T_D^{gq}}^{\text{elig}}$$

by the corresponding eight lawful classes:

$$\begin{aligned} \sigma_0(r_{\text{anc}}) &= \rho_{\text{anc}}, & \sigma_0(r_{\text{ten}}) &= \rho_{\text{ten}}, & \sigma_0(r_{\text{skin}}) &= \rho_{\text{skin}}, & \sigma_0(r_{\text{GR}}) &= \rho_{\text{GR}}, \\ \sigma_0(r_{\text{QM}}) &= \rho_{\text{QM}}, & \sigma_0(r_{\text{bdry}}) &= \rho_{\text{bdry}}, & \sigma_0(r_{\text{trans}}) &= \rho_{\text{trans}}, & \sigma_0(r_{\text{match}}) &= \rho_{\text{match}}. \end{aligned}$$

Theorem 21.3 (Assignment eligibility and compatibility). *The map σ_0 is injective, pointwise eligible, and globally compatible:*

$$\forall r \in \mathcal{R}_D^{gq} \text{ Elig}_D(\sigma_0(r), r), \quad \text{Compat}_D(\sigma_0) = 1.$$

Proof. The residue classes lie in disjoint sort summands, so the map is injective. Pointwise eligibility is theorem 21.1. The comparison isomorphisms from the accepted presentations to the target vertices make every generating square of Θ_D^{ABC} commute by theorem 11.4. $\square$

Corollary 21.4 (Nonempty compatible occupation attempt). *The pointwise eligible compatible injection space is nonempty.*

22 Independent live-residue formation

Definition 22.1 (Pre-compatible injection space). Let

$$\Sigma_{D,\text{pre}}^{gg} = \left\{ \tau : \mathcal{R}_D^{gg} \hookrightarrow \text{Residue}_{T_D^{gg}}^{\text{elig}} : \forall r \text{ Elig}_D(\tau(r), r), \text{Compat}_D(\tau) = 1 \right\}.$$

The live residue is formed independently by

$$\text{Residue}_{T_D^{gg}}^{\text{live}} = \bigcup_{\tau \in \Sigma_{D,\text{pre}}^{gg}} \text{im}(\tau).$$

Lemma 22.2 (Pointwise eligible injection rigidity). *Every pointwise eligible injection $\tau : \mathcal{R}_D^{gg} \hookrightarrow \text{Residue}_{T_D^{gg}}^{\text{elig}}$ equals σ_0 .*

Proof. For each role, the target fibre contains exactly one eligible residue class by theorem 21.1. Therefore $\tau(r) = \sigma_0(r)$ for every role. This uses target-fibre finality, not equality with chosen presentation names. $\square$

Theorem 22.3 (Pre-compatible assignment uniqueness).

$$\Sigma_{D,\text{pre}}^{gg} = \{\sigma_0\}.$$

Proof. The displayed assignment lies in $\Sigma_{D,\text{pre}}^{gg}$ by theorem 21.3; every pointwise eligible injection equals it by lemma 22.2. $\square$

Theorem 22.4 (Live-residue identification).

$$\text{Residue}_{T_D^{gg}}^{\text{live}} = \text{im}(\sigma_0) = \{\rho_{\text{anc}}, \rho_{\text{ten}}, \rho_{\text{skin}}, \rho_{\text{GR}}, \rho_{\text{QM}}, \rho_{\text{bdry}}, \rho_{\text{trans}}, \rho_{\text{match}}\}.$$

Proof. The live residue is the union of the images of all compatible complete injections. The injection space is singleton by theorem 22.3. $\square$

Corollary 22.5 (No unused live residue). *Every live target-relative residue is used, and no additional live residue remains for same-certificate selection.*

23 Role-realization completeness of the certificate

Theorem 23.1 (Certificate role-realization completeness). *The fixed certificate C_D^{ABC} is role-realization complete for the frozen target T_D^{gg} .*

Proof. We verify the seven clauses of the imported completeness definition.

Candidate completeness. Theorem 8.3 and corollary 8.4 prove coverage of every response generated by the fixed certificate, while external moduli changes are transparently reset.

Quotient completeness. Presentation equivalence is lawful isomorphism in a target presentation groupoid. The explicit tensor-colimit construction and dependent skin theorem prove those two fibres nonempty and connected; the other six uniform fibre-finality theorems complete the list. Proposition 9.6 retains nonisomorphic standing-bearing differences.

Role-set completeness. Proposition 5.27 fixes the eight roles, generic interfaces, indexing category, and target diagram before residue inspection.

Eligibility completeness. Theorem 10.6 separates generic interface work, target fidelity, and incidence typing without using canonical presentation equality.

Compatibility completeness. Theorem 11.3 derives the exact generating edge registry from the certificate operations, and definition 11.2 requires commutation of every target-diagram square or declared directed-exchange relation.

Live-residue completeness. Theorem 22.4 forms the live residue from all complete compatible injections and proves that every target-relative eligible class is used.

Failure completeness. Theorem 12.2 separates malformed generation, external target parameters, interface failure, target-fidelity failure, quotient failure, compatibility failure, live-residue failure, orbit failure, and explicit reset.

All seven conditions hold. $\square$

Corollary 23.2 (Cardinality shadow).

$$|\mathcal{R}_D^{gg}| = |\text{Residue}_{T_D^{gg}}^{\text{live}}| = 8.$$

Proof. The equality follows from the bijective target assignment after the stronger completeness and compatibility results. It is not used as their substitute. $\square$

24 Certificate-lawful role-incidence automorphisms

Definition 24.1 (Complete invariant tuple). Let

$$\mathbb{I}_D^{gg} = (C_D^{ABC}, T_D^{gg}, \Theta_D^{ABC}, \mathcal{R}_D^{gg}, \text{Residue}_{T_D^{gg}}^{\text{live}}, \text{Elig}_D, \text{Compat}_D, \text{St}, \text{Ten}, \text{Bd}, \text{Cont}, \text{Fail}, \simeq_{\text{law}}, \text{Report})$$

be the complete assignment invariant tuple. It includes the generic sort signatures, frozen target parameters, lawful presentation groupoids, target-relative eligibility, the full generating incidence registry, continuation/exchange data, failure registry, and report status.

Definition 24.2 (Lawful role-incidence automorphism). A lawful automorphism is a pair

$$\gamma = (\alpha, \beta), \quad \alpha : \mathcal{R}_D^{gg} \rightarrow \mathcal{R}_D^{gg}, \quad \beta : \text{Residue}_{T_D^{gg}}^{\text{live}} \rightarrow \text{Residue}_{T_D^{gg}}^{\text{live}},$$

of bijections preserving the full tuple $\mathbb{I}_D^{gg}$, including every sort signature, target parameter, eligibility edge, target-diagram incidence, standing-bearing invariant, continuation, failure, lawful-equivalence, and report coordinate. The group is Γ_D^{lawful} .

Lemma 24.3 (Role rigidity). *Every lawful automorphism fixes each role.*

Proof. The eight roles carry pairwise nonisomorphic generic interface operations: anchor tuple, colimit ledger, kernel congruence, Lorentzian Cauchy development, projective self-adjoint process, quotient product, Moore-history object, and matching cone. Preserving the sort signatures forces $\alpha = \text{id}_{\mathcal{R}_D^{gg}}$. $\square$

Lemma 24.4 (Residue rigidity). *Every lawful automorphism fixes each live residue.*

Proof. Role rigidity fixes the role endpoint of every eligibility edge. Each role has one target-relative eligible live class by theorem 21.1. Since β preserves eligibility and the target parameters, it fixes each class. $\square$

Theorem 24.5 (Automorphism-group theorem).

$$\Gamma_D^{\text{lawful}} = \{1\}.$$

Proof. Every lawful automorphism has identity role and residue components by lemmas 24.3 and 24.4. The identity preserves the complete tuple and is therefore the unique group element. $\square$

Remark 24.6 (Internal symmetries are already quotiented). Gauge, coordinate, basis, phase, clock-presentation, record-label, and other internal presentation symmetries act inside the target presentation groupoids and are removed before Γ_D^{lawful} is formed. The group here acts on target-relative residue classes and their incidence diagram.

25 Unique complete assignment orbit

Definition 25.1 (Final assignment space).

$$\Sigma_D^{gg} = \left\{ \sigma : \mathcal{R}_D^{gg} \xrightarrow{\sim} \text{Residue}_{T_D^{gg}}^{\text{live}} : \forall r \text{ Elig}_D(\sigma(r), r), \text{Compat}_D(\sigma) = 1 \right\}.$$

Theorem 25.2 (Literal assignment uniqueness).

$$\Sigma_D^{gg} = \{\sigma_0\}.$$

Proof. The displayed assignment is a compatible bijection by theorems 21.3 and 22.4. Every final assignment is a pointwise eligible injection and therefore equals σ_0 by lemma 22.2. $\square$

Theorem 25.3 (Unique complete assignment orbit).

$$\left| \Sigma_D^{gg} / \Gamma_D^{\text{lawful}} \right| = 1.$$

Proof. The final assignment space is singleton and the lawful automorphism group is trivial. $\square$

Corollary 25.4 (No selector gap). *The fixed certificate contains no inequivalent complete compatible assignment requiring a further selector.*

26 Role-Occupancy Closure and licensed realization

Theorem 26.1 (Role-Occupancy Closure for the capstone certificate). *The fixed pair (C_D^{ABC}, T_D^{gg}) satisfies Role-Occupancy Closure:*

$$\text{ROC}(C_D^{ABC}, T_D^{gg}).$$

Proof. Role-Occupancy Closure is the statement that the compatible bijective assignment space over the independently formed live target-relative residue has one orbit under certificate-lawful role-incidence automorphisms. This is theorem 25.3. $\square$

Theorem 26.2 (Complete realization licensure). *The fixed certificate licenses one complete constraint-native realization class of the frozen target, unique up to lawful equivalence:*

$$\text{RealizationLicensed}(C_D^{ABC}, T_D^{gg}).$$

Proof. The certificate is role-realization complete by theorem 23.1 and satisfies Role-Occupancy Closure by theorem 26.1. The sufficiency direction of the imported theorem applies. Every candidate response generated by the certificate is classified, presentation multiplicity is lawfully quotiented, every forced role has a target-faithful occupant, all generating incidence squares commute, every live residue is used, and every failed or reset route is explicit. $\square$

Corollary 26.3 (Certificate-relative uniqueness only). *The theorem is relative to the exact fixed certificate and target-parameter diagram. Changing a source, initial-value package, potential, initial ray, RPI seed, matching map, or richer-scope operation changes the certificate and requires a new candidate/residue analysis.*

Part V

Finite Audit Artifact and Separating Models

27 Finite target-diagram audit object

The analytic proof is represented by a finite audit object only after the target-fibred candidate, quotient, eligibility, compatibility, and live-residue theorems have been established. The finite object does not manufacture those results.

Definition 27.1 (Finite certificate data). The finite audit certificate is

$$\mathfrak{C}_D^{\text{fin}} = (R, X, \text{type}, \text{TargetParam}, E, J, \text{InvHash}, \text{Fail}, \text{Report}),$$

where:

1. R is the eight-element forced role set;
2. X is the eight-element target-relative live residue;
3. type records the pairwise distinct role fingerprints;
4. TargetParam records the source-locked target-parameter hash at each vertex;
5. $E \subseteq X \times R$ records the proved target-relative eligibility relation;
6. J is the exact twenty-five-edge compatibility registry;
7. InvHash records the lawful residue-class invariant hashes;
8. Fail records unresolved failure status;
9. Report records the scope intersection.

Proposition 27.2 (Finite certificate correspondence). *The finite audit object is the image of the mathematical target diagram, eligible residue, live residue, and lawful automorphism action under the finite identifier functor. It contains one target-relative residue per role, every required generating edge, no unresolved failure, and the displayed assignment σ_0 .*

Proof. The role, target-parameter, residue, eligibility, edge, and failure ledgers are generated from the theorem statements and exact source locks after the analytic fibre theorems are complete. No finite identifier is used as a premise in a target-fibre finality proof. $\square$

Example 27.3 (Generic moduli versus the finite target fibre). A second potential may exist in the generic quantum moduli. It does not appear as an extra residue in this finite object because the object records candidates generated by the fixed certificate and the fibre over its frozen target parameter. A reset certificate for the second potential would have a different target-parameter hash and would require a new finite audit object.

28 Separating failure models

Example 28.1 (Generic interface multiplicity is preserved). Let $Q_1, Q_2 \in \mathbf{M}_{\mathbf{QM}}$ be self-adjoint projective quantum systems satisfying the same generic interface burden, but carrying different fixed potentials or initial rays. Both may satisfy

$$\mathbf{IElig}_D(Q_i, r_{\mathbf{QM}}).$$

If $\theta_{\mathbf{QM}} = Q_1$ and no lawful invariant-interface isomorphism identifies the full process and fixed data of Q_2 with Q_1 , then Q_2 fails $\mathbf{TFid}_D(-, r_{\mathbf{QM}})$ for the present target point. It is retained as a legitimate point of the generic moduli space and determines a different target instance; it is not discarded as an “ineligible theory.”

Example 28.2 (Two presentations, one target residue). Let P and P' be gauge, basis, or chart presentations of one fixed target parameter θ_r , with lawful interface isomorphisms

$$u : P \xrightarrow{\sim} \theta_r, \quad u' : P' \xrightarrow{\sim} \theta_r.$$

Then $(u')^{-1}u$ is a lawful presentation isomorphism over θ_r . Raw presentation multiplicity increases while the target-relative residue class does not.

Example 28.3 (Missing required incidence). Delete the tensor-to-GR source-ancestry incidence from an otherwise canonical certificate. Every local role candidate may remain interface-eligible and target-faithful, but the tuple does not present the frozen diagram because one generating comparison square is absent. The tuple is pointwise eligible and globally incompatible.

Example 28.4 (Excess target-faithful residue). Suppose a malformed certificate declares two target-faithful GR presentations to be distinct residue classes despite a lawful isomorphism over $\theta_{\mathbf{GR}}$. Presentation-quotient completeness fails before live-residue formation. Retaining the false split would create excess live residue and prevent a complete bijective occupation.

Example 28.5 (Lawful symmetry versus physical alternative). A toy target may have two interchangeable roles and two residue classes related by an automorphism of the complete incidence tuple. Two literal assignments then lie in one lawful orbit. The capstone roles are differently typed, but

internal presentation stabilizers may still be nontrivial; the theorem requires one orbit, not a trivial group by definition.

Example 28.6 (Explicit reset). Adding a new interacting curved-spacetime QFT source, a different initial state, or a nonzero metric–quantum cross-response changes a vertex or arrow of Θ_D^{ABC} . The response grammar classifies the change as an explicit reset. It defines a new target parameter diagram and requires a new completion theorem; it is not unused residue in the present certificate.

Proposition 28.7 (Separating power). *The preceding examples independently separate generic interface eligibility, exact target fidelity, presentation quotienting, global compatibility, live-residue coverage, lawful automorphism quotienting, and target reset.*

29 Bounded finite-certificate checker

Definition 29.1 (Software verification scope). The checker verifies only the finite image of the accepted mathematical results:

1. the exact ordered role registry and target-base identifiers;
2. the exact covariant tensor schema A01–A25;
3. the fifteen source-locked generator cores, twenty-four source-locked overlap cores, and their exact forward-cosieve completions;
4. every generator and overlap incidence map required by completed support;
5. one composable primary-to-target path witness for every completed component;
6. every completed overlap embedding component and its naturality/joint-monicity flags;
7. the sixteen source-locked probes and their unique completed supports, plus nine joint-probe components;
8. the finite tensor-colimit/dependent-skin attestation registry;
9. generic-interface status, target-fidelity status, presentation classes, and explicit candidate failures;
10. exact presence, uniqueness, endpoint typing, relation label, and pass status of all twenty-five target compatibility edges;
11. lawful presentation-link connectivity, eligible injections, live residue, bijective final assignments, and representative-orbit count;
12. all seven certificate-completeness flags and absence of unresolved failures.

Nonclaim 29.2 (Bounded software). The checker does not prove the response-grammar theorem, existence of source-locked core data, left Kan universal properties, joint-image relation theorem, tensor-colimit universal property, probe completeness, skin maximality, target locking, Cauchy uniqueness, Wigner–Stone theory, record or compatibility finality, accepted Modules A–C, or physical correctness. It consumes their finite certified outputs.

Proposition 29.3 (Software correspondence). *For a JSON certificate satisfying Appendix H, the checker returns PASS if and only if the finite support-completion, incidence-map, path-provenance,*

overlap-embedding, probe, target-fibred candidate, compatibility, residue, assignment, and representative-orbit data satisfy the exact frozen schema and declared expected counts.

Proof. The program first validates the exact role and target-base registries, target-diagram hash, co-variant variance convention, schema arrows, generator and overlap identifiers, primary and completed supports, completion methods, cosieve closure, incidence-map totality, path witnesses, overlap embedding components, probe support completions, joint probes, construction flags, and target edges. It then classifies candidates, computes target-eligible presentation classes, checks lawful presentation-link connectivity, enumerates every injective role-to-class assignment, filters declared finite incompatibilities, forms the union of assignment images, retains bijections onto that live union, and computes representative orbit counts from the certified presentation components. Finally it verifies all seven completeness flags, the empty unresolved-failure list, and the expected live-residue, assignment, and orbit counts. $\square$

The regression suite includes missing or extra completed support targets; non-cosieve supports; wrong primary supports or completion methods; missing, duplicate, or endpoint-mismatched incidence maps; incomplete path provenance; malformed overlap embeddings; missing probes or construction flags; deletion, duplication, or alteration of target edges; malformed roles and bases; missing occupants; hidden-load-only fibres; false presentation splits; disconnected lawful representatives; unresolved failures; and alternate-target candidates.

Part VI

Capstone Handoff and Adversarial Closure

30 Standalone main theorem package

Theorem 30.1 (Target-Fibred Eight-Fibre Exhaustion Theorem). *Let Θ_D^{ABC} be the completed target-parameter diagram obtained from the source-locked direct interfaces, the explicitly constructed tensor colimit, and the dependent skin kernel. For each role r , the generic interface moduli may contain many objects, but the target presentation groupoid over the point θ_r is nonempty and connected. Consequently:*

$$\begin{aligned} \text{Residue}_{\text{anc}}^{\text{elig}} &= \{\rho_{\text{anc}}\}, & \text{Residue}_{\text{ten}}^{\text{elig}} &= \{\rho_{\text{ten}}\}, & \text{Residue}_{\text{skin}}^{\text{elig}} &= \{\rho_{\text{skin}}\}, & \text{Residue}_{\text{GR}}^{\text{elig}} &= \{\rho_{\text{GR}}\}, \\ \text{Residue}_{\text{QM}}^{\text{elig}} &= \{\rho_{\text{QM}}\}, & \text{Residue}_{\text{bdry}}^{\text{elig}} &= \{\rho_{\text{bdry}}\}, & \text{Residue}_{\text{trans}}^{\text{elig}} &= \{\rho_{\text{trans}}\}, & \text{Residue}_{\text{match}}^{\text{elig}} &= \{\rho_{\text{match}}\}. \end{aligned}$$

Proof. Tensor and skin nonemptiness/finality follow from theorems 5.17, 5.24 and 14.3 and corollary 15.5; apply the remaining six uniform fibre-finality theorems to their frozen target points. $\square$

Theorem 30.2 (Generated-Candidate and Exact Live-Residue Theorem). *Every target-relevant candidate response generated by C_D^{ABC} is represented by the response grammar, and the independently*

formed live residue is exactly the eight target-relative lawful classes.

Proof. Candidate coverage is theorem 8.3 and corollary 8.4. Exact live-residue identification is theorem 22.4. $\square$

Theorem 30.3 (Unique Role-Occupancy and Realization Theorem). *The certificate is role-realization complete and*

$$\left| \Sigma_D^{gq} / \Gamma_D^{\text{lawful}} \right| = 1.$$

Hence one complete constraint-native realization class of the frozen target is licensed, unique up to lawful equivalence.

Proof. Combine theorems 23.1, 25.3 and 26.2. $\square$

The theorem is a fibre theorem over the fixed certificate parameter diagram. It does not say that all points of the GR, quantum, transport, or matching moduli are equivalent. Replacing a source, potential, initial ray, initial data, RPI seed, or matching package changes the target parameter and triggers a new certificate analysis.

31 Capstone handoff

Module D closes the two gates remaining after accepted Modules A–C.

Gate	Corrected Module D handoff
G1	Source-locked tensor generator coverage; explicit tensor colimit and induced joint probe; dependent skin kernel; certificate-generated candidate completeness; target-lock and target-fibred eligibility; eight fibre-finality theorems; exact compatibility diagram; live-residue closure; complete certificate inhabitation.
G7	Complete invariant tuple; lawful role-incidence action; literal assignment uniqueness; one assignment orbit; Role-Occupancy licensure.

Proposition 31.1 (Integration-safe theorem-use package). *A capstone integration may import:*

1. *the source-locked tensor-colimit, induced-probe, dependent-skin, target-lock, and certificate-generated candidate-completeness theorems;*
2. *the eight uniform and target-relative fibre-finality theorems;*
3. *exact target-diagram compatibility and live-residue formation;*
4. *role-realization completeness;*
5. *the lawful automorphism group and unique assignment orbit;*
6. *Role-Occupancy and realization licensure for the fixed target.*

It may not infer uniqueness across different certificate parameter diagrams or new physical dynamics.

32 Adversarial map and standing filters

Objection	Standing?	Corrected response
The roles were chosen to match the occupants.	Standing if true.	The role types, generic moduli, indexing category, and target diagram are fixed before candidate formation.
The candidate class excludes alternative potentials by definition.	Standing if arbitrary envelope objects are called candidates.	Candidate responses are exactly those generated by the fixed certificate, as required by Role-Occupancy Closure. Alternative potentials remain legitimate points of the generic moduli and create reset certificates.
Eligibility requires the canonical source, state, or seed.	Standing if true.	Generic interface eligibility is separated from target fidelity. Exact values are base parameters of the independently frozen target diagram, not names or values inserted into the role burden.
The quotient collapses whatever the probes do not see.	Standing.	Presentation equivalence is lawful interface isomorphism over a fixed target parameter. Probe profiles support local finality but do not define the quotient.
Cauchy uniqueness identifies different source theories.	No; not used.	The GR theorem is uniform over fixed Cauchy parameters and does not identify different base points.
Wigner–Stone makes all Hamiltonians equivalent.	No; not used.	The quantum theorem is uniform over a fixed projective process, domain, potential, and initial ray. Different base points remain distinct.
Local flow uniqueness identifies different RPI seeds.	No; not used.	The transport theorem fixes a seed parameter first and proves uniqueness of its presentations and flow.
The tensor colimit is assumed canonical.	Standing if only a cocone or an already assembled ledger is supplied.	The source-locked generator diagram is antecedent. The total ledger is constructed sort-wise as a tagged coproduct modulo the least declared-overlap congruence; exact overlap, the universal mediator, and mediator uniqueness are proved before the tensor target is defined.
The skin kernel depends on an unproved complete tensor probe.	Standing if tensor completion is open.	The tensor probe and tensor-to-downstream maps are uniquely induced by the proved colimit. The joint probe is formed afterward and proved standing-complete before its kernel pair is used.
The live residue was defined to be canonical.	No.	It is defined as the union over all compatible complete injections and identified only after target-fibre finality and compatibility are proved.
The compatibility list is incomplete.	Standing if a generated cross-sort operation is missing.	The exact twenty-five generating arrows are derived from the certificate operations; all composites are paths in the indexing category.

Objection	Standing?	Corrected response
The software accepts an empty edge registry.	Closed.	The corrected checker enforces exact required edge identifiers, endpoints, relations, allowed pairs, uniqueness, and pass status, with deletion regressions.
The theorem proves all quantum-gravity dynamics unique.	No such claim.	It is uniqueness of one fixed certificate-parameter diagram; other moduli points require new certificates.

Theorem 32.1 (Adversarial-map gate). *No final verdict on Module D is permitted until every objection has been tested against the generated-candidate theorem, target-lock theorem, target-fibred eligibility, local uniform fibre theorem, complete incidence diagram, failure registry, and Role-Occupancy dependency chain.*

33 Conclusion

The first Module D release treated broad same-envelope interface objects as candidate residues and then attempted to force the exact accepted source, state, initial data, seed, and matching values through eligibility. That conflated generic interface moduli with the target-relative candidate residue. The corrected construction separates them.

First, the generic moduli retain different potentials, source systems, initial states, initial data, RPI seeds, and matching packages. Second, the exact accepted Modules A–C outputs fix the direct interfaces and a source-locked tensor generator diagram before any candidate is inspected. Module D then constructs the tensor target as the tagged coproduct modulo the least declared-overlap congruence, proves its universal property, induces every tensor incidence and probe, and only then constructs the skin target as the maximal kernel pair of the completed joint probe. Third, the candidate set is the closure of the responses actually generated by the fixed certificate: coordinate presentations, lawful redescriptions, restrictions, licensed quotients, incidences, failures, reports, and explicit resets. Role-interface eligibility is uniform in the base parameter; target fidelity then selects the fibre over the frozen target point without naming the canonical presentation.

Presentation equivalence is lawful interface isomorphism over that point, not equality under a discretionary probe list. The eight local theorems are uniform fibre theorems: explicit colimit construction and colimit uniqueness for the tensor ledger, dependent standing-complete kernel-pair finality for skin, Cauchy uniqueness for fixed GR/source data, Wigner–Stone uniqueness for a fixed projective process, final record and compatibility quotients, flow uniqueness for a fixed RPI seed, and naturality/cubical uniqueness for fixed matching data. Specialization to the accepted target diagram gives one lawful class in each of the eight role fibres while leaving the generic moduli untouched.

The complete twenty-five-edge incidence diagram makes the accepted assignment globally compatible. The live residue is formed from all complete compatible injections, not declared in advance. Each role has one target-relative class, so the live residue is exactly the eight accepted classes. The role-incidence automorphism group preserves the pairwise distinct sort signatures and frozen target parameters and is therefore trivial. The final assignment space has one lawful orbit.

Thus:

$$\boxed{\text{RealizationLicensed}(C_D^{ABC}, T_D^{gg}) \quad \text{and} \quad \left| \Sigma_D^{gg} / \Gamma_D^{\text{lawful}} \right| = 1.}$$

This is the exact certificate-relative closure required by FEP-1.0. It is not uniqueness of all GR, QM, or quantum-gravity models; a changed source, potential, state, seed, or matching object changes the target-parameter diagram and must be audited under a new certificate.

A Formal predicate package

Definition A.1 (Target diagram and generated candidate data). For the fixed certificate and target, use the covariant physical-incidence convention $X : \mathcal{S}_D \rightarrow \mathbf{Set}$, so every registry arrow $u : s' \rightarrow s$ acts as $X(u) : X(s') \rightarrow X(s)$, and first form the source-locked antecedent diagram

$$\mathbb{G}_D : \mathbf{K}_D^{\text{ten}} \longrightarrow \mathbf{TenInt}_D,$$

construct its colimit

$$L_D^0 = \text{colim } \mathbb{G}_D,$$

and induce the unique tensor probe

$$p_{\text{ten}} : L_D^0 \longrightarrow \mathcal{V}_D.$$

The complete joint probe is then formed, and the skin object is its lawful kernel pair:

$$S_D^0 = \text{KerPair}(\text{Probe}_{\Theta_D^{ABC}}).$$

Only after these constructions is the complete target diagram defined:

$$\Theta_D^{ABC} : \mathbf{J}_D^{gq} \rightarrow \mathbf{Int}, \quad \theta_r = \Theta_D^{ABC}(r).$$

The candidate set is

$$\text{Cand}_D^{ABC} = \{\llbracket t \rrbracket : t \in \mathfrak{G}_D^{ABC}, t \text{ non-reset}\}.$$

For each role, $\pi_r : \mathcal{P}_r \rightarrow \mathbf{M}_r$ is the presentation fibration.

Definition A.2 (Generic and target-relative eligibility). For a generated presentation P of sort r ,

$$\text{IElig}_D(P, r) \iff \text{CompleteProv}_C(P) \wedge \text{Ifc}_r(P; \pi_r(P)) \wedge \text{NoHiddenLoad}_r(P).$$

Target fidelity is

$$\text{TFid}_D(P, r) \iff \pi_r(P) \cong \theta_r$$

through an incident-compatible lawful invariant isomorphism. For $x = [P]_{\sim_r}$,

$$\text{Elig}_D(x, r) \iff \text{IElig}_D(P, r) \wedge \text{TFid}_D(P, r) \wedge \text{IncType}_r(P).$$

Definition A.3 (Lawful presentation quotient).

$$P \sim_r P' \iff P, P' \text{ lie in the same connected component of } \mathcal{P}_r[\theta_r].$$

The residue is

$$\text{Residue}(C_D^{ABC}) = \bigsqcup_{r \in \mathcal{R}_D^{gq}} \text{Ob}(\mathcal{F}_r^{\text{gen}}) / \sim_r.$$

Definition A.4 (Compatibility). For a pointwise eligible injection τ , choose target-fidelity isomor-

phisms λ_r . Then

$$\text{Compat}_D(\tau) = 1$$

iff every generating square of J_D^{gq} commutes or satisfies its declared directed-exchange relation, no local failure remains, and the report status is the accepted scope intersection.

Definition A.5 (Pre-compatible assignments and live residue).

$$\Sigma_{D,\text{pre}}^{gq} = \{\tau : \mathcal{R}_D^{gq} \hookrightarrow \text{Residue}_{T_D^{gq}}^{\text{elig}} : \forall r \text{ Elig}_D(\tau(r), r), \text{Compat}_D(\tau) = 1\},$$

$$\text{Residue}_{T_D^{gq}}^{\text{live}} = \bigcup_{\tau \in \Sigma_{D,\text{pre}}^{gq}} \text{im}(\tau).$$

Definition A.6 (Final assignments and lawful automorphisms).

$$\Sigma_D^{gq} = \{\sigma : \mathcal{R}_D^{gq} \xrightarrow{\sim} \text{Residue}_{T_D^{gq}}^{\text{live}} : \forall r \text{ Elig}_D(\sigma(r), r), \text{Compat}_D(\sigma) = 1\}.$$

The lawful group is

$$\Gamma_D^{\text{lawful}} = \text{Aut}(\mathbb{I}_D^{gq}),$$

with

$$(\gamma \cdot \sigma)(r) = \beta(\sigma(\alpha^{-1}r)).$$

Theorem A.7 (Formal endpoint).

$$\Sigma_{D,\text{pre}}^{gq} = \Sigma_D^{gq} = \{\sigma_0\}, \quad \text{Residue}_{T_D^{gq}}^{\text{live}} = \text{im}(\sigma_0), \quad \Gamma_D^{\text{lawful}} = \{1\}.$$

Therefore

$$|\Sigma_D^{gq} / \Gamma_D^{\text{lawful}}| = 1.$$

B Target-fibred invariant ledger

Fibre	Generic base parameter	Uniform finality mechanism	Frozen target point
Anchor	Domain, target, regularity, boundary, clock, scale, operator-domain, scope tuple.	Presentations of one fixed tuple differ by lawful encoding only.	FEP/source-lock tuple.
Tensor	Source-locked standing-bearing generator diagram and declared overlap maps.	Tagged-coproduct quotient, universal mediator, and colimit uniqueness.	Constructed colimit of the accepted RPI/AER/A-C generator diagram.

Fibre	Generic base parameter	Uniform finality mechanism	Frozen target point
Skin	Joint lawful standing-bearing probe formed after tensor completion.	Standing-complete induced probe and uniqueness of its maximal kernel congruence.	Kernel pair of the completed frozen joint probe.
GR/source	Source orbit, field system, gauge, regularity, initial/boundary data, exchange, interval.	Source ancestry plus local gauge-reduced Cauchy uniqueness for fixed data.	Accepted Module A/C metric-source package.
QM	Projective carrier/process, generator/domain, potential, initial ray, boundary, closure.	Wigner–Stone and Schrödinger representation uniqueness for the fixed process/data.	Accepted Module B/C quantum package.
Boundary	Reuse, joint-witness/no-transport profiles, boundary certificates, descended processes.	Final record and compatibility quotient universal properties.	Accepted record/compatibility package.
Transport	RPI chart, clock, seed, standing/failure boundary, common incidence, strict/open status.	Local flow uniqueness and duration-transport equivalence for the fixed seed package.	Accepted Module C history package.
Matching	Common charge line, source/target functors, incidence restrictions, exchange cells, cube, report.	Natural-intertwiner, exchange-action, scalar-matching, and cubical uniqueness for fixed endpoints.	Accepted Modules B–C matching package.

Remark B.1. The generic base columns may contain multiple inequivalent points. Module D specializes the uniform finality theorem in each row to the source-locked frozen point; it does not identify different rows or different base points in one row.

C Cross-coordinate target-diagram registry

ID	From	To	Required target incidence
E01	anchor	tensor	Common target and envelope.
E02	anchor	skin	Probe kernel formed on the same target.
E03	anchor	GR	Frozen Lorentzian/Cauchy/source envelope.
E04	anchor	QM	Frozen Hilbert/domain/clock envelope.
E05	anchor	boundary	Fixed record and compatibility scopes.
E06	anchor	transport	Fixed RPI domain, clock, and failure envelope.

ID	From	To	Required target incidence
E07	anchor	matching	Matching preserves report and calibration envelope.
E08	tensor	skin	Every standing-bearing generator factors through the final skin quotient.
E09	tensor	GR	GR source is the accepted Module A source in the tensor ledger.
E10	tensor	QM	Operator/source and inertial data agree with Module B ancestry.
E11	tensor	boundary	Boundary witnesses are the tensor-ledger witnesses.
E12	tensor	transport	History, failure, and incidence data agree with Module C.
E13	tensor	matching	Source/operator ancestry and common charge agree.
E14	transport	GR	Metric process descends from the accepted upstream history.
E15	transport	QM	Quantum process descends from that same history.
E16	transport	boundary	Record and compatibility processes descend from that history.
E17	transport	matching	Intertwiners and cube use the same history and incidence.
E18	GR	matching	Metric intertwiner and active readout use the fixed GR/source object.
E19	QM	matching	Quantum intertwiner and inertial readout use the fixed QM object.
E20	boundary	matching	Boundary intertwiners use the final quotient objects.
E21	skin	GR	GR presentation changes factor through lawful skin.
E22	skin	QM	Basis/phase/domain-core changes factor through lawful skin.
E23	skin	boundary	Record/witness relabelling factors through lawful skin.
E24	skin	transport	RPI/clock-presentation changes factor through lawful skin.
E25	skin	matching	Natural-isomorphism and cube relabelling factor through lawful skin.

Proposition C.1. *The registry is the generating-arrow set of J_D^{gg} . Compatibility of all composites follows by path functoriality and the accepted Module C cubical coherence relations.*

D Failure registry

Code	Locus	Diagnosis
F01	Target lock	Proposed object changes a frozen target parameter; new certificate required.
F02	Generation	Response is not produced by a licensed certificate constructor.
F03	Candidate	Missing coordinate operation or malformed generated interface.
F04	Sort	Wrong role fingerprint.
F05	Interface	Generic role burden fails.
F06	Target fidelity	Candidate lies over a different moduli point.
F07	Quotient	Lawful presentation duplicate retained.
F08	Quotient	Nonisomorphic standing-bearing candidates collapsed.
F09	Compatibility	One or more required target-diagram squares fail.
F10	Exchange	Directed exchange source, target, incidence, or cocycle fails.
F11	Live residue	Pointwise eligible class occurs in no complete compatible injection.
F12	Live residue	Excess live class survives unused.
F13	Assignment	Missing role, repeated residue, or non-bijective coverage.
F14	Orbit	More than one lawful assignment orbit remains.
F15	Repair	Empirical, semantic, probabilistic, or post-hoc selector invoked.

E Complete target-fibred certificate schema

The completed abstract certificate parameter diagram is

$$\Theta_D^{ABC} = (\theta_{\text{anc}}, \theta_{\text{ten}}, \theta_{\text{skin}}, \theta_{\text{GR}}, \theta_{\text{QM}}, \theta_{\text{bdry}}, \theta_{\text{trans}}, \theta_{\text{match}}; \theta_e)_{e \in E01-E25}.$$

The direct imported vertices and tensor generator diagram are antecedent. The tensor vertex is the proved colimit of that generator diagram, and the skin vertex is the kernel pair of the joint probe formed after tensor completion. Choose one lawful presentation of each completed vertex to display

$$\mathcal{C}_{D,0}^{gg} = (C_{\text{anc}}^0, C_{\text{ten}}^0, C_{\text{skin}}^0, C_{\text{GR}}^0, C_{\text{QM}}^0, C_{\text{bdry}}^0, C_{\text{trans}}^0, C_{\text{match}}^0).$$

The abstract target points, not these representative objects, are fixed before candidate inspection.

Role	Frozen invariant target point	Construction or accepted presentation source
r_{anc}	FEP and source-lock envelope	FEP/source locks
r_{ten}	Proved colimit of the source-locked standing-bearing generator diagram	Explicit Module D colimit construction from RPI and Modules A–C
r_{skin}	Maximal lawful kernel of the joint probe induced after tensor completion	ATS, constructed tensor probe, and accepted final probes
r_{GR}	Fixed source/Cauchy/process invariant package	Gravity, Einstein, Modules A,C
r_{QM}	Fixed projective process/domain/potential/state package	Schrödinger, Modules B,C
r_{bdry}	Fixed final quotient/process package	Record, compatibility, Module C
r_{trans}	Fixed RPI seed/history/incidence package	Module C
r_{match}	Fixed charge/intertwiner/exchange/cube package	Modules B–C

F Theorem-use ledger

Imported theorem	Exact Module D use	Prohibited promotion
Kernel/ATS	Target identity, no back-flow, eight work types	No generation of candidates or physics
RPI	Fixed upstream target object and response generators	No claim that RPI alone gives certificate closure
Module A	Source generators, probes, overlap maps, and exact accepted source vertex/ancestry	No generic uniqueness of all sources
Module B	Operator/source, charge, form-factor, contraction, matching generators and overlaps; exact accepted mass-match vertex	No massless, unstable, radiating, or arbitrary-sector extension
Module C	History, record, compatibility, incidence, exchange, cube generators and overlaps; exact accepted continuation vertex	No generic uniqueness of all RPI seeds or histories

Imported theorem	Exact Module D use	Prohibited promotion
Einstein	Classification and local uniqueness after fixed source/gauge/initial/boundary data	No identification of different source systems or initial data
Schrödinger/ Wigner/Stone	Implementation of one fixed projective process and fixed domain/potential/ray	No identification of different Hamiltonians, potentials, or states
Record and compatibility	Final quotients for fixed profiles	No hidden transport or outcome selection
Role-Occupancy Closure	Realization iff one assignment orbit under a complete certificate	No generation of missing occupants and no use before completeness

G Claim-status matrix

Claim	Status	Boundary
Source-locked tensor generator diagram	Proved here from imported hashes and overlap registry	Antecedent to tensor and skin construction.
Constructed tensor colimit	Proved here by explicit coproduct/quotient and universal property	Exact accepted Modules A–C generator diagram.
Completed joint probe and skin kernel	Proved here after tensor completion	Frozen target only.
Frozen target-parameter diagram	Proved here from direct source locks plus tensor/skin universal constructions	Exact accepted Modules A–C instance.
Certificate-generated candidate completeness	Proved here	Responses generated by the fixed certificate, not all objects in generic moduli.
Lawful presentation-quotient completeness	Proved here	Target fibres and actual interface isomorphisms only.
Eight fibre-finality theorems	Proved here	One class per fixed base parameter in accepted envelopes.

Claim	Status	Boundary
Generic GR/QM/transport moduli singleton	Not claimed	Different parameters require different certificates.
Complete target-diagram compatibility	Proved here	Twenty-five generating incidences and composites.
Exact live residue	Proved here	Target-relative residue only.
Lawful automorphism group	Proved here	After presentation quotienting and target locking.
Unique assignment orbit	Proved here	Fixed certificate and completed target diagram.
Role-Occupancy and realization licensure	Proved here	Certificate-relative realization class.
New interaction dynamics or all QG uniqueness	Not claimed	Outside the frozen certificate.
Strong equivalence or empirical parameters	Not claimed	Outside Module D.

H Finite software schema

The bounded JSON artifact uses the following target-fibred and support-completed schema:

```
{
  "module": "AASC-GRQM-MD-OCC-1.0",
  "schema_version": "MD-OCC-1.4",
  "release": "v1.4.0-cosieve-support-corrected",
  "tensor_interface_category": "Set(S_D)/W_D",
  "tensor_schema_variance": "covariant_source_to_target",
  "tensor_variance_convention": "covariant:Set(S_D)",
  "roles": ["anc", "ten", "skin", "gr", "qm", "bdry", "trans", "match"],
  "required_tensor_schema_arrow_ids": ["A01", "...", "A25"],
  "tensor_schema_arrows": [
    {
      "id": "A01",
      "source_sort": "S02",
      "target_sort": "S01",
      "operation": "source_attachment_to_upstream_response",
      "variance": "covariant",
      "present": true
    }
  ],
  "tensor_generator_supports": [
    {
      "id": "G03",
      "primary_support": ["S02", "S03", "S15"],
      "completed_support": ["S01", "...", "S15"],
      "completion_method": "left_Kan_incidence_completion",
      "cosieve_closed": true
    }
  ],
  "tensor_overlap_supports": [
    {
      "id": "002",
      "left_generator": "G02",
      "right_generator": "G03",
      "primary_support": ["S02"],
      "completed_support": ["S01", "S02"],
      "completion_method": "left_Kan_joint_image_completion",
      "cosieve_closed": true,
      "jointly_monoid": true
    }
  ],
}
```

```

"tensor_generator_incidence_maps": [
  {"generator_id":"G03","arrow_id":"A01",
   "source_sort":"S02","target_sort":"S01","registered":true}
],
"tensor_overlap_incidence_maps": [
  {"overlap_id":"002","arrow_id":"A01",
   "source_sort":"S02","target_sort":"S01","registered":true}
],
"tensor_completion_witness_paths": [
  {"object_kind":"generator","object_id":"G03",
   "origin_sort":"S02","target_sort":"S01","path_arrows":"A01"}
],
"tensor_overlap_embeddings": [
  {"overlap_id":"002","side":"left","generator_id":"G02",
   "sort_id":"S01","natural":true,"jointly_mononic":true}
],
"tensor_probe_supports": [
  {"id":"P03","generator_id":"G03","core_probe_sorts":["S03","S15"],
   "completed_support":["S01","...", "S15"],
   "extension_mode":"unique_left_Kan_probe_extension","present":true}
],
"tensor_colimit_flags": {"...":true},
"candidates": [{"...": "..."}],
"compatibility_edges": [{"id":"E01","...": "...", "pass":true}],
"completeness": {"candidate_complete":true,"...":true},
"unresolved_failures": [],
"expected_live_residue": ["rho_anc","...", "rho_match"],
"expected_final_assignments": 1,
"expected_orbit_count": 1
}

```

The checker hard-codes the source-to-target variance, exact arrow specifications, primary supports, forward closures, completion methods, required maps, path composability, adjacent-generator containment, overlap embedding components, probe support completion, construction flags, and target-edge registry. The finite ledgers are audit witnesses only; they do not substitute for the manuscript's left-Kan, image-relation, colimit, induced-probe, kernel-pair, or physical-interface proofs. Empty, incomplete, duplicated, extra, malformed, non-cosieve, endpoint-mismatched, or failed registries are rejected before assignment enumeration.

I Independent hostile-referee checklist

A referee should verify:

1. the generic interface moduli are distinguished from the frozen target-parameter diagram;
2. the direct target interfaces and tensor generator diagram are source locked before candidate inspection;
3. each accepted package first forms a source-locked core diagram on a full core subcategory;
4. every total generator is its left Kan extension and every completed support is a forward-closed

- cosieve;
5. every required completed incidence map and primary-to-target witness path is constructed;
 6. every completed overlap is the natural joint-image relation in the adjacent completed generator product;
 7. the tensor target is constructed only afterward as the tagged coproduct modulo the least declared-overlap congruence;
 8. the exact-overlap theorem excludes undeclared collapse and extra generator load;
 9. the universal mediator exists and is unique for every lawful cocone;
 10. tensor-to-downstream incidences and the tensor probe are induced by the colimit universal property;
 11. the joint probe and skin kernel are formed only after tensor completion;
 12. candidate completeness concerns responses generated by the fixed certificate, as in the imported Role-Occupancy theorem;
 13. generic interface eligibility is independent of canonical presentation values;
 14. target fidelity compares with abstract invariant target points rather than presentation names;
 15. presentation equivalence is lawful interface isomorphism, not equality under an assumed-complete probe list;
 16. each fibre theorem is uniform over fixed base parameters and does not identify different moduli points;
 17. all twenty-five generating compatibility edges are present and every composite follows functorially;
 18. the live residue is formed from all compatible injections before identification;
 19. the automorphism group preserves the target diagram and does not erase standing-bearing differences;
 20. the occupancy theorem is invoked only after seven-part certificate completeness;
 21. software claims remain finite and enforce the exact tensor and edge registries;
 22. the final result is certificate-relative.

J Symbol index

Symbol	Meaning
C_D^{ABC}	Fixed capstone certificate after accepted Modules A–C.
T_D^{gq}	Frozen gravity–quantum capstone target.
$\mathcal{R}_D^{gq}$	Eight forced certificate roles.

Symbol	Meaning
J_D^{gq}	Finite target-incidence indexing category.
Θ_D^{ABC}	Frozen abstract target-parameter diagram.
S_D	Finite tensor-incidence schema.
TenInt_D	Slice $\mathbf{Set}^{S_D}/\mathcal{W}_D$ of covariant physical-incidence diagrams.
$\mathbb{G}_D$	Source-locked tensor generator/overlap diagram.
L_D^0	Explicitly constructed colimit tensor ledger.
p_{ten}	Tensor probe induced by the colimit universal property.
S_D^0	Dependent lawful kernel-pair skin object.
M_r	Generic interface moduli groupoid for role r .
θ_r	Frozen target point in M_r .
$\mathfrak{G}_D^{ABC}$	Certificate response grammar.
Cand_D^{ABC}	Candidate responses generated by the fixed certificate.
$\pi_r : \mathcal{P}_r \rightarrow M_r$	Presentation fibration.
$\text{Ifc}_r(P; b)$	Generic role-interface burden over parameter b .
IElig_D	Generic interface eligibility.
TFid_D	Target fidelity to θ_r .
$\sim_r$	Lawful target-presentation groupoid equivalence.
ρ_r^0	Target-relative lawful residue class.
Elig_D	Complete target-relative role eligibility.
Compat_D	Target-diagram compatibility predicate.
$\Sigma_{D,\text{pre}}^{gq}$	Pointwise eligible compatible injections.
$\text{Residue}_{T_D^{gq}}^{\text{live}}$	Union of images of all pre-compatible injections.
Σ_D^{gq}	Compatible bijections onto the live residue.
Γ_D^{lawful}	Certificate-lawful role-incidence automorphism group.
σ_0	Displayed accepted target assignment.

K Source-locked core interfaces, cosieve completions, and probe ledger

This appendix is the publication-facing shadow of the machine-readable registries:

- `tensor_schema.csv` and `tensor_schema_arrows.csv`;
- `tensor_generators.csv` and `tensor_generator_incidence_maps.csv`;
- `tensor_overlaps.csv` and `tensor_overlap_incidence_maps.csv`;
- `tensor_support_completion.csv`;
- `tensor_support_closure_audit.csv`;
- `tensor_support_completion_certificate.json`;

- `tensor_completion_paths.csv` and `tensor_completion_witness_paths.csv`;
- `tensor_generator_incidence_maps.csv` and `tensor_overlap_incidence_maps.csv`;
- `tensor_overlap_embeddings.csv`;
- `tensor_probe_registry.csv` and `joint_probe_registry.csv`;
- `tensor_colimit_certificate.json`.

The accepted modules supply the core diagrams. Module D constructs their lawful covariant completions before forming the tensor colimit. Newly supported values are incidence images forced by the frozen schema; they are not additional independent tensor generators.

Support-domain lock. Each source package has a source-locked core support. Its lawful generator is the left Kan extension along the source-locked primary incidence inclusion. Each overlap is the joint image relation of the completed core overlap in the product of its completed generators over the frozen witness diagram. The resulting nonempty supports are forward-closed cosieves, and every outgoing arrow has a registered map.

K.1 Schema incidence operations

Every row $A_{ij} : S_{\text{source}} \rightarrow S_{\text{target}}$ acts covariantly as $X(A_{ij}) : X(S_{\text{source}}) \rightarrow X(S_{\text{target}})$.

ID	Source	Target	Lawful incidence operation
A01	S02	S01	source attachment to upstream response.
A02	S03	S02	ward regular to attached source.
A03	S04	S02	composite ledger to attached source.
A04	S05	S02	normalized ray to concrete source.
A05	S06	S05	operator emt to diagonal source family.
A06	S07	S06	charge formfactor to operator emt.
A07	S08	S07	mass readout to common charge.
A08	S09	S01	history to upstream response.
A09	S10	S02	metric process to accepted source.
A10	S10	S09	metric process to upstream history.
A11	S11	S06	quantum process to attached operator.
A12	S11	S09	quantum process to upstream history.
A13	S12	S09	record process to upstream history.
A14	S13	S09	compatibility process to upstream history.
A15	S14	S09	incidence cube to upstream history.
A16	S14	S10	incidence cube to metric restriction.
A17	S14	S11	incidence cube to quantum restriction.
A18	S14	S12	incidence cube to record restriction.
A19	S14	S13	incidence cube to compatibility restriction.

ID	Source	Target	Lawful incidence operation
A20	S14	S08	incidence cube to mass matching.
A21	S15	S03	failure report to ward status.
A22	S15	S04	failure report to composite status.
A23	S15	S09	failure report to history status.
A24	S15	S14	failure report to cube status.
A25	S16	S15	overlap registry to report status.

K.2 Generator primary supports and completed cosieves

The primary column records the native source-locked sorts. The completed column is the exact forward closure in the frozen schema. “Maps” is the number of registered arrow maps whose source lies in the completed support.

ID	Generator	Core support	Completed support	No.
G01	RPI/AER re- sponse	S01, S09	S01, S09	1
G02	Module A at- tachment	S01, S02	S01, S02	1
G03	Module A Ward	S02, S03, S15	S01, S02, S03, S04, S05, S06, S07, S08, S09, S10, S11, S12, S13, S14, S15	24
G04	Module A composite	S02, S04, S15	S01, S02, S03, S04, S05, S06, S07, S08, S09, S10, S11, S12, S13, S14, S15	24
G05	Module B ray source	S02, S05	S01, S02, S05	2
G06	Module B op- erator EMT	S05, S06	S01, S02, S05, S06	3
G07	Module B charge/form factor	S06, S07	S01, S02, S05, S06, S07	4
G08	Module B mass readout	S07, S08, S14	S01, S02, S05, S06, S07, S08, S09, S10, S11, S12, S13, S14	18
G09	Module C his- tory	S01, S09, S15	S01, S02, S03, S04, S05, S06, S07, S08, S09, S10, S11, S12, S13, S14, S15	24
G10	Module C met- ric	S02, S09, S10, S14	S01, S02, S05, S06, S07, S08, S09, S10, S11, S12, S13, S14	18
G11	Module C quantum	S06, S09, S11, S14	S01, S02, S05, S06, S07, S08, S09, S10, S11, S12, S13, S14	18

ID	Generator	Core support	Completed support	No.
G12	Module C record	S09, S12, S14	S01, S02, S05, S06, S07, S08, S09, S10, S11, S12, S13, S14	18
G13	Module C compatibility	S09, S13, S14	S01, S02, S05, S06, S07, S08, S09, S10, S11, S12, S13, S14	18
G14	Module C cube	S08, S09, S10, S11, S12, S13, S14, S15	S01, S02, S03, S04, S05, S06, S07, S08, S09, S10, S11, S12, S13, S14, S15	24
G15	Capstone fail- ure/report	S03, S04, S09, S14, S15, S16	S01, S02, S03, S04, S05, S06, S07, S08, S09, S10, S11, S12, S13, S14, S15, S16	25

For the source-locked primary incidence inclusion $j_k : \mathcal{C}_k \hookrightarrow \mathcal{S}_D$, the completed value at t is

$$\overline{G}_k(t) = \operatorname{colim}_{(s,p) \in (j_k \downarrow t)} G_k^\circ(s),$$

and $u : t \rightarrow t'$ acts by $[s, p, x] \mapsto [s, u \circ p, x]$. This formula supplies the actual set and map at every newly required target sort. The 222 individual registrations are in `tensor_generator_incidence_maps.csv`.

K.3 Primary overlap relations and completed supports

The core overlap is the common interface supplied by the adjacent accepted packages. Its left Kan extension is mapped to the product of the two completed generators over the witness diagram; the completed overlap is the objectwise image of that joint map.

ID	Left	Right	Primary	Completed support	Maps
O01	G01	G02	S01	S01	0
O02	G02	G03	S02	S01, S02	1
O03	G02	G04	S02	S01, S02	1
O04	G02	G05	S02	S01, S02	1
O05	G05	G06	S05	S01, S02, S05	2
O06	G06	G07	S06	S01, S02, S05, S06	3
O07	G07	G08	S07	S01, S02, S05, S06, S07	4
O08	G01	G09	S01, S09	S01, S09	1
O09	G02	G10	S02	S01, S02	1
O10	G09	G10	S09	S01, S09	1
O11	G06	G11	S06	S01, S02, S05, S06	3
O12	G09	G11	S09	S01, S09	1
O13	G09	G12	S09	S01, S09	1
O14	G09	G13	S09	S01, S09	1
O15	G09	G14	S09	S01, S09	1

ID	Left	Right	Primary	Completed support	Maps
O16	G10	G14	S10, S14	S01, S02, S05, S06, S07, S08, S09, S10, S11, S12, S13, S14	18
O17	G11	G14	S11, S14	S01, S02, S05, S06, S07, S08, S09, S10, S11, S12, S13, S14	18
O18	G12	G14	S12, S14	S01, S02, S05, S06, S07, S08, S09, S10, S11, S12, S13, S14	18
O19	G13	G14	S13, S14	S01, S02, S05, S06, S07, S08, S09, S10, S11, S12, S13, S14	18
O20	G08	G14	S08, S14	S01, S02, S05, S06, S07, S08, S09, S10, S11, S12, S13, S14	18
O21	G03	G15	S15	S01, S02, S03, S04, S05, S06, S07, S08, S09, S10, S11, S12, S13, S14, S15	24
O22	G04	G15	S15	S01, S02, S03, S04, S05, S06, S07, S08, S09, S10, S11, S12, S13, S14, S15	24
O23	G09	G15	S15	S01, S02, S03, S04, S05, S06, S07, S08, S09, S10, S11, S12, S13, S14, S15	24
O24	G14	G15	S15	S01, S02, S03, S04, S05, S06, S07, S08, S09, S10, S11, S12, S13, S14, S15	24

The 208 overlap-incidence maps are registered in `tensor_overlap_incidence_maps.csv`. The combined completion has 430 explicit generator/overlap arrow maps. The support audit checks all 975 object-arrow pairs and contains no supported-source/unsupported-target case.

K.4 Declared overlap meanings

ID	Left	Right	Source-locked common interface
O01	G01	G02	Upstream response identity carried into Module A source attachment.
O02	G02	G03	Attached source class shared with Ward regularity and boundary data.
O03	G02	G04	Attached source ancestry shared with completed composite terms.
O04	G02	G05	Accepted Module A source presentations shared with normalized-ray family.
O05	G05	G06	Normalized-ray source family shared with polarization operator reconstruction.
O06	G06	G07	Attached operator EMT shared with charge and form-factor interface.
O07	G07	G08	Common charge shared with active and inertial readouts.

ID	Left	Right	Source-locked common interface
O08	G01	G09	RPI standing-bearing response carried by upstream history.
O09	G02	G10	Accepted Module A source shared with metric process.
O10	G09	G10	Metric process descended from the same upstream history.
O11	G06	G11	Attached operator EMT shared with quantum process.
O12	G09	G11	Quantum process descended from the same upstream history.
O13	G09	G12	Record process descended from the same upstream history.
O14	G09	G13	Compatibility process descended from the same upstream history.
O15	G09	G14	Common incidence belongs to the same upstream history.
O16	G10	G14	Metric exchange restriction and cube incidence.
O17	G11	G14	Quantum exchange restriction and cube incidence.
O18	G12	G14	Record exchange restriction and cube incidence.
O19	G13	G14	Compatibility exchange restriction and cube incidence.
O20	G08	G14	Common mass-charge matching enters the continuation cube.
O21	G03	G15	Ward boundary failure and report status.
O22	G04	G15	Composite closure failure and report status.
O23	G09	G15	History standing failure and no-repair status.
O24	G14	G15	Exchange cube failure and reportability status.

The completed overlap relation is, by construction, the objectwise joint image in the fibre product of the two completed generators; its joint map is monic. Individual projections need not remain monic after lawful forgetting. The tensor colimit identifies only pairs in this completed declared relation and then takes the least generated equivalence.

K.5 Probe completion

Each core generator probe is extended by the same Kan adjunction used for its generator. The completed probe evaluates a class $[s, p, x]$ by transporting the accepted core probe value along p . Core overlap compatibility extends uniquely to the completed overlap relation.

ID	Generator	Core probe sorts	Standing burden
P01	G01	S01, S09	RPI and AER response identity and standing profile.
P02	G02	S02	Metric-response attachment and source distribution profile.
P03	G03	S03, S15	Support Ward regularity boundary and exchange profile.

ID	Generator	Core probe sorts	Standing burden
P04	G04	S04, S15	Composite term coverage channel balance and external-current profile.
P05	G05	S05	Normalized-ray concrete source profile.
P06	G06	S06	Operator EMT matrix-element and source-attachment profile.
P07	G07	S07	Poincare charge Casimir and zero-transfer form-factor profile.
P08	G08	S08, S14	Active inertial calibration and naturality profile.
P09	G09	S09, S15	Upstream history clock standing failure and report profile.
P10	G10	S10, S14	Metric Cauchy and exchange profile.
P11	G11	S11, S14	Full-ray process generator-domain and exchange profile.
P12	G12	S12, S14	Record quotient lineage reuse and exchange profile.
P13	G13	S13, S14	Compatibility quotient no-transport and exchange profile.
P14	G14	S14, S15	Common incidence cocycle cube and report profile.
P15	G15	S15	Frozen failure and reportability profile.
P16	G15	S16	Declared overlap and incidence registry profile.

K.6 Finite support-completion certificate

The certificate records fifteen generator completions, twenty-four overlap completions, 222 generator maps, 208 overlap maps, and the following checked conditions:

1. every core diagram and core overlap is source locked;
2. every completed support is the exact forward closure of its core;
3. every supported source has a supported target;
4. every required arrow map is registered;
5. the canonical unit maps retain primary data modulo only accepted primary incidence relations;
6. completed probes and overlap compatibility are induced by adjunction;
7. the completion adds only forced incidence images and no independent generator.

These finite ledgers certify support-domain typing. They do not substitute for the left-Kan, image-relation, tensor-colimit, probe, or skin universal-property proofs.

References

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